

# EDA RED WINE QUALITY

## 1. Định nghĩa vấn đề (Define Problem)



Wine Dataset

| Chỉ số                      | Ý nghĩa ngắn gọn                                                            |
|-----------------------------|-----------------------------------------------------------------------------|
| <b>Fixed acidity</b>        | Axit không bay hơi (malic, lactic, tartaric...), đặc trưng cơ bản của rượu. |
| <b>Volatile acidity</b>     | Axit bay hơi (chủ yếu acetic). Quá cao gây mùi giấm.                        |
| <b>Citric acid</b>          | Axit hữu cơ nhẹ, tạo độ "tươi" cho rượu.                                    |
| <b>Residual sugar</b>       | Đường còn lại sau lên men (g/L), quyết định độ ngọt.                        |
| <b>Chlorides</b>            | Hàm lượng muối (ion $\text{Cl}^-$ ) trong rượu.                             |
| <b>Free sulfur dioxide</b>  | $\text{SO}_2$ tự do, có tác dụng kháng khuẩn & chống oxi hóa.               |
| <b>Total sulfur dioxide</b> | Tổng $\text{SO}_2$ (tự do + liên kết).                                      |
| <b>Density</b>              | Mật độ rượu so với nước, phụ thuộc cồn & đường.                             |
| <b>pH</b>                   | Độ axit/kiềm (0–14), rượu thường 3–4.                                       |
| <b>Sulphates</b>            | Phụ gia góp phần vào mức $\text{SO}_2$ , bảo quản rượu.                     |
| <b>Alcohol</b>              | % cồn của rượu.                                                             |
| <b>Quality</b>              | Điểm đánh giá chất lượng cảm quan (3–8).                                    |

## 2. Chuẩn bị vấn đề (Prepare Problem)

### 2.1. Khai báo thư viện (Load Libraries)

```
In [58]: import pandas as pd #for data manipulation operations
import seaborn as sns #for data visualization operations
import matplotlib.pyplot as plt #for data visualization operations
```

### 2.2. Nạp dữ liệu (Load Dataset)

```
In [59]: wine = pd.read_csv("winequality-red.csv")
df = wine.copy()
df.head(n = 10)
```

Out[59]:

|   | fixed<br>acidity | volatile<br>acidity | citric<br>acid | residual<br>sugar | chlorides | free<br>sulfur<br>dioxide | total<br>sulfur<br>dioxide | density | pH   | sulphates | a |
|---|------------------|---------------------|----------------|-------------------|-----------|---------------------------|----------------------------|---------|------|-----------|---|
| 0 | 7.4              | 0.70                | 0.00           | 1.9               | 0.076     | 11.0                      | 34.0                       | 0.9978  | 3.51 | 0.56      |   |
| 1 | 7.8              | 0.88                | 0.00           | 2.6               | 0.098     | 25.0                      | 67.0                       | 0.9968  | 3.20 | 0.68      |   |
| 2 | 7.8              | 0.76                | 0.04           | 2.3               | 0.092     | 15.0                      | 54.0                       | 0.9970  | 3.26 | 0.65      |   |
| 3 | 11.2             | 0.28                | 0.56           | 1.9               | 0.075     | 17.0                      | 60.0                       | 0.9980  | 3.16 | 0.58      |   |
| 4 | 7.4              | 0.70                | 0.00           | 1.9               | 0.076     | 11.0                      | 34.0                       | 0.9978  | 3.51 | 0.56      |   |
| 5 | 7.4              | 0.66                | 0.00           | 1.8               | 0.075     | 13.0                      | 40.0                       | 0.9978  | 3.51 | 0.56      |   |
| 6 | 7.9              | 0.60                | 0.06           | 1.6               | 0.069     | 15.0                      | 59.0                       | 0.9964  | 3.30 | 0.46      |   |
| 7 | 7.3              | 0.65                | 0.00           | 1.2               | 0.065     | 15.0                      | 21.0                       | 0.9946  | 3.39 | 0.47      |   |
| 8 | 7.8              | 0.58                | 0.02           | 2.0               | 0.073     | 9.0                       | 18.0                       | 0.9968  | 3.36 | 0.57      |   |
| 9 | 7.5              | 0.50                | 0.36           | 6.1               | 0.071     | 17.0                      | 102.0                      | 0.9978  | 3.35 | 0.80      |   |

2.3. Thông tin cơ bản

```
In [60]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   fixed acidity         1599 non-null   float64
1   volatile acidity      1599 non-null   float64
2   citric acid           1599 non-null   float64
3   residual sugar        1599 non-null   float64
4   chlorides             1599 non-null   float64
5   free sulfur dioxide   1599 non-null   float64
6   total sulfur dioxide  1599 non-null   float64
7   density               1599 non-null   float64
8   pH                   1599 non-null   float64
9   sulphates             1599 non-null   float64
10  alcohol               1599 non-null   float64
11  quality               1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```

- Dataset size:** 1599 dòng, 12 cột
- Data types:** Tất cả biến đều dạng số (numeric)
- Missing values:** Không có giá trị null

## 3. Làm sạch, xem xét

### 3.1. Thống kê mô tả (Descriptive Statistics)

#### (1) Hiện thị một số thông tin về dữ liệu

```
In [61]: # shape
print(f'+ Shape: {df.shape}')

# types
print(f'+ Data Types: \n{df.dtypes}')

# head, tail
print(f'+ Contents: ')
display(df.head(5))
display(df.tail(5))

# info
df.info()
```

+ Shape: (1599, 12)

+ Data Types:

```
fixed acidity      float64
volatile acidity   float64
citric acid        float64
residual sugar     float64
chlorides          float64
free sulfur dioxide float64
total sulfur dioxide float64
density            float64
pH                 float64
sulphates          float64
alcohol            float64
quality            int64
```

dtype: object

+ Contents:

|   | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides | free sulfur dioxide | total sulfur dioxide | density | pH   | sulphates | alc |
|---|---------------|------------------|-------------|----------------|-----------|---------------------|----------------------|---------|------|-----------|-----|
| 0 | 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.9978  | 3.51 | 0.56      |     |
| 1 | 7.8           | 0.88             | 0.00        | 2.6            | 0.098     | 25.0                | 67.0                 | 0.9968  | 3.20 | 0.68      |     |
| 2 | 7.8           | 0.76             | 0.04        | 2.3            | 0.092     | 15.0                | 54.0                 | 0.9970  | 3.26 | 0.65      |     |
| 3 | 11.2          | 0.28             | 0.56        | 1.9            | 0.075     | 17.0                | 60.0                 | 0.9980  | 3.16 | 0.58      |     |
| 4 | 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11.0                | 34.0                 | 0.9978  | 3.51 | 0.56      |     |

|             | fixed<br>acidity | volatile<br>acidity | citric<br>acid | residual<br>sugar | chlorides | free<br>sulfur<br>dioxide | total<br>sulfur<br>dioxide | density | pH   | sulphates |
|-------------|------------------|---------------------|----------------|-------------------|-----------|---------------------------|----------------------------|---------|------|-----------|
| <b>1594</b> | 6.2              | 0.600               | 0.08           | 2.0               | 0.090     | 32.0                      | 44.0                       | 0.99490 | 3.45 | 0.58      |
| <b>1595</b> | 5.9              | 0.550               | 0.10           | 2.2               | 0.062     | 39.0                      | 51.0                       | 0.99512 | 3.52 | 0.76      |
| <b>1596</b> | 6.3              | 0.510               | 0.13           | 2.3               | 0.076     | 29.0                      | 40.0                       | 0.99574 | 3.42 | 0.75      |
| <b>1597</b> | 5.9              | 0.645               | 0.12           | 2.0               | 0.075     | 32.0                      | 44.0                       | 0.99547 | 3.57 | 0.71      |
| <b>1598</b> | 6.0              | 0.310               | 0.47           | 3.6               | 0.067     | 18.0                      | 42.0                       | 0.99549 | 3.39 | 0.66      |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   fixed acidity          1599 non-null   float64
1   volatile acidity       1599 non-null   float64
2   citric acid            1599 non-null   float64
3   residual sugar         1599 non-null   float64
4   chlorides              1599 non-null   float64
5   free sulfur dioxide    1599 non-null   float64
6   total sulfur dioxide   1599 non-null   float64
7   density                1599 non-null   float64
8   pH                    1599 non-null   float64
9   sulphates              1599 non-null   float64
10  alcohol                1599 non-null   float64
11  quality                1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```

## (2) Kiểm tra tính toàn vẹn của dữ liệu

```
In [62]: has_null = df.isnull().sum().any()
has_nan  = df.isna().sum().any()
n_duplicated = df.duplicated().sum()
print(f'Tính toàn vẹn dữ liệu:')
print(f'+ Có giá trị Null: {has_null}')
if has_null:
    display.display(df[df.isnull().any(axis=1)])
print(f'+ Có giá trị Nan: {has_nan}')
if has_nan:
    display.display(df[df.isna().any(axis=1)])
print(f'+ Số dòng trùng: {n_duplicated}')
```

Tính toàn vẹn dữ liệu:  
 + Có giá trị Null: False  
 + Có giá trị Nan: False  
 + Số dòng trùng: 240

### Xử lý trùng lặp dữ liệu



```
In [63]: df.drop_duplicates(keep='first', inplace=True)
```

```
In [64]: has_null = df.isnull().sum().any()
has_nan = df.isna().sum().any()
n_duplicated = df.duplicated().sum()
print(f'Tính toàn vẹn dữ liệu:')
print(f'+ Có giá trị Null: {has_null}')
if has_null:
    display.display(df[df.isnull().any(axis=1)])
print(f'+ Có giá trị Nan: {has_nan}')
if has_nan:
    display.display(df[df.isna().any(axis=1)])
print(f'+ Số dòng trùng: {n_duplicated}')
```

Tính toàn vẹn dữ liệu:  
+ Có giá trị Null: False  
+ Có giá trị Nan: False  
+ Số dòng trùng: 0

### Đổi tên cột để dễ truy vấn

```
In [65]: df.rename(columns = {"fixed acidity": "fixed_acidity", "volatile acidity": "volatil",
                             "citric acid": "citric_acid", "residual sugar": "residual_sugar",
                             "chlorides": "chlorides", "free sulfur dioxide": "free_sulfur_d",
                             "total sulfur dioxide": "total_sulfur_dioxide"}, inplace = True)
```

```
In [66]: display(df)
```

|           | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-----------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>0</b>  | 7.4           | 0.700            | 0.00        | 1.90           | 0.076     | 11.0                |
| <b>1</b>  | 7.8           | 0.880            | 0.00        | 2.60           | 0.098     | 25.0                |
| <b>2</b>  | 7.8           | 0.760            | 0.04        | 2.30           | 0.092     | 15.0                |
| <b>3</b>  | 11.2          | 0.280            | 0.56        | 1.90           | 0.075     | 17.0                |
| <b>5</b>  | 7.4           | 0.660            | 0.00        | 1.80           | 0.075     | 13.0                |
| <b>6</b>  | 7.9           | 0.600            | 0.06        | 1.60           | 0.069     | 15.0                |
| <b>7</b>  | 7.3           | 0.650            | 0.00        | 1.20           | 0.065     | 15.0                |
| <b>8</b>  | 7.8           | 0.580            | 0.02        | 2.00           | 0.073     | 9.0                 |
| <b>9</b>  | 7.5           | 0.500            | 0.36        | 6.10           | 0.071     | 17.0                |
| <b>10</b> | 6.7           | 0.580            | 0.08        | 1.80           | 0.097     | 15.0                |
| <b>12</b> | 5.6           | 0.615            | 0.00        | 1.60           | 0.089     | 16.0                |
| <b>13</b> | 7.8           | 0.610            | 0.29        | 1.60           | 0.114     | 9.0                 |
| <b>14</b> | 8.9           | 0.620            | 0.18        | 3.80           | 0.176     | 52.0                |
| <b>15</b> | 8.9           | 0.620            | 0.19        | 3.90           | 0.170     | 51.0                |
| <b>16</b> | 8.5           | 0.280            | 0.56        | 1.80           | 0.092     | 35.0                |
| <b>17</b> | 8.1           | 0.560            | 0.28        | 1.70           | 0.368     | 16.0                |
| <b>18</b> | 7.4           | 0.590            | 0.08        | 4.40           | 0.086     | 6.0                 |
| <b>19</b> | 7.9           | 0.320            | 0.51        | 1.80           | 0.341     | 17.0                |
| <b>20</b> | 8.9           | 0.220            | 0.48        | 1.80           | 0.077     | 29.0                |
| <b>21</b> | 7.6           | 0.390            | 0.31        | 2.30           | 0.082     | 23.0                |
| <b>22</b> | 7.9           | 0.430            | 0.21        | 1.60           | 0.106     | 10.0                |
| <b>23</b> | 8.5           | 0.490            | 0.11        | 2.30           | 0.084     | 9.0                 |
| <b>24</b> | 6.9           | 0.400            | 0.14        | 2.40           | 0.085     | 21.0                |
| <b>25</b> | 6.3           | 0.390            | 0.16        | 1.40           | 0.080     | 11.0                |
| <b>26</b> | 7.6           | 0.410            | 0.24        | 1.80           | 0.080     | 4.0                 |
| <b>28</b> | 7.1           | 0.710            | 0.00        | 1.90           | 0.080     | 14.0                |
| <b>29</b> | 7.8           | 0.645            | 0.00        | 2.00           | 0.082     | 8.0                 |
| <b>30</b> | 6.7           | 0.675            | 0.07        | 2.40           | 0.089     | 17.0                |
| <b>31</b> | 6.9           | 0.685            | 0.00        | 2.50           | 0.105     | 22.0                |
| <b>32</b> | 8.3           | 0.655            | 0.12        | 2.30           | 0.083     | 15.0                |

|           | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-----------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>33</b> | 6.9           | 0.605            | 0.12        | 10.70          | 0.073     | 40.0                |
| <b>34</b> | 5.2           | 0.320            | 0.25        | 1.80           | 0.103     | 13.0                |
| <b>35</b> | 7.8           | 0.645            | 0.00        | 5.50           | 0.086     | 5.0                 |
| <b>36</b> | 7.8           | 0.600            | 0.14        | 2.40           | 0.086     | 3.0                 |
| <b>37</b> | 8.1           | 0.380            | 0.28        | 2.10           | 0.066     | 13.0                |
| <b>38</b> | 5.7           | 1.130            | 0.09        | 1.50           | 0.172     | 7.0                 |
| <b>39</b> | 7.3           | 0.450            | 0.36        | 5.90           | 0.074     | 12.0                |
| <b>41</b> | 8.8           | 0.610            | 0.30        | 2.80           | 0.088     | 17.0                |
| <b>42</b> | 7.5           | 0.490            | 0.20        | 2.60           | 0.332     | 8.0                 |
| <b>43</b> | 8.1           | 0.660            | 0.22        | 2.20           | 0.069     | 9.0                 |
| <b>44</b> | 6.8           | 0.670            | 0.02        | 1.80           | 0.050     | 5.0                 |
| <b>45</b> | 4.6           | 0.520            | 0.15        | 2.10           | 0.054     | 8.0                 |
| <b>46</b> | 7.7           | 0.935            | 0.43        | 2.20           | 0.114     | 22.0                |
| <b>47</b> | 8.7           | 0.290            | 0.52        | 1.60           | 0.113     | 12.0                |
| <b>48</b> | 6.4           | 0.400            | 0.23        | 1.60           | 0.066     | 5.0                 |
| <b>49</b> | 5.6           | 0.310            | 0.37        | 1.40           | 0.074     | 12.0                |
| <b>50</b> | 8.8           | 0.660            | 0.26        | 1.70           | 0.074     | 4.0                 |
| <b>51</b> | 6.6           | 0.520            | 0.04        | 2.20           | 0.069     | 8.0                 |
| <b>52</b> | 6.6           | 0.500            | 0.04        | 2.10           | 0.068     | 6.0                 |
| <b>53</b> | 8.6           | 0.380            | 0.36        | 3.00           | 0.081     | 30.0                |
| <b>54</b> | 7.6           | 0.510            | 0.15        | 2.80           | 0.110     | 33.0                |
| <b>55</b> | 7.7           | 0.620            | 0.04        | 3.80           | 0.084     | 25.0                |
| <b>56</b> | 10.2          | 0.420            | 0.57        | 3.40           | 0.070     | 4.0                 |
| <b>57</b> | 7.5           | 0.630            | 0.12        | 5.10           | 0.111     | 50.0                |
| <b>58</b> | 7.8           | 0.590            | 0.18        | 2.30           | 0.076     | 17.0                |
| <b>59</b> | 7.3           | 0.390            | 0.31        | 2.40           | 0.074     | 9.0                 |
| <b>60</b> | 8.8           | 0.400            | 0.40        | 2.20           | 0.079     | 19.0                |
| <b>61</b> | 7.7           | 0.690            | 0.49        | 1.80           | 0.115     | 20.0                |
| <b>62</b> | 7.5           | 0.520            | 0.16        | 1.90           | 0.085     | 12.0                |
| <b>63</b> | 7.0           | 0.735            | 0.05        | 2.00           | 0.081     | 13.0                |

|           | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-----------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>64</b> | 7.2           | 0.725            | 0.05        | 4.65           | 0.086     | 4.0                 |
| <b>66</b> | 7.5           | 0.520            | 0.11        | 1.50           | 0.079     | 11.0                |
| <b>67</b> | 6.6           | 0.705            | 0.07        | 1.60           | 0.076     | 6.0                 |
| <b>68</b> | 9.3           | 0.320            | 0.57        | 2.00           | 0.074     | 27.0                |
| <b>69</b> | 8.0           | 0.705            | 0.05        | 1.90           | 0.074     | 8.0                 |
| <b>70</b> | 7.7           | 0.630            | 0.08        | 1.90           | 0.076     | 15.0                |
| <b>71</b> | 7.7           | 0.670            | 0.23        | 2.10           | 0.088     | 17.0                |
| <b>72</b> | 7.7           | 0.690            | 0.22        | 1.90           | 0.084     | 18.0                |
| <b>73</b> | 8.3           | 0.675            | 0.26        | 2.10           | 0.084     | 11.0                |
| <b>74</b> | 9.7           | 0.320            | 0.54        | 2.50           | 0.094     | 28.0                |
| <b>75</b> | 8.8           | 0.410            | 0.64        | 2.20           | 0.093     | 9.0                 |
| <b>77</b> | 6.8           | 0.785            | 0.00        | 2.40           | 0.104     | 14.0                |
| <b>78</b> | 6.7           | 0.750            | 0.12        | 2.00           | 0.086     | 12.0                |
| <b>79</b> | 8.3           | 0.625            | 0.20        | 1.50           | 0.080     | 27.0                |
| <b>80</b> | 6.2           | 0.450            | 0.20        | 1.60           | 0.069     | 3.0                 |
| <b>81</b> | 7.8           | 0.430            | 0.70        | 1.90           | 0.464     | 22.0                |
| <b>82</b> | 7.4           | 0.500            | 0.47        | 2.00           | 0.086     | 21.0                |
| <b>83</b> | 7.3           | 0.670            | 0.26        | 1.80           | 0.401     | 16.0                |
| <b>84</b> | 6.3           | 0.300            | 0.48        | 1.80           | 0.069     | 18.0                |
| <b>85</b> | 6.9           | 0.550            | 0.15        | 2.20           | 0.076     | 19.0                |
| <b>86</b> | 8.6           | 0.490            | 0.28        | 1.90           | 0.110     | 20.0                |
| <b>87</b> | 7.7           | 0.490            | 0.26        | 1.90           | 0.062     | 9.0                 |
| <b>88</b> | 9.3           | 0.390            | 0.44        | 2.10           | 0.107     | 34.0                |
| <b>89</b> | 7.0           | 0.620            | 0.08        | 1.80           | 0.076     | 8.0                 |
| <b>90</b> | 7.9           | 0.520            | 0.26        | 1.90           | 0.079     | 42.0                |
| <b>92</b> | 8.6           | 0.490            | 0.29        | 2.00           | 0.110     | 19.0                |
| <b>94</b> | 5.0           | 1.020            | 0.04        | 1.40           | 0.045     | 41.0                |
| <b>95</b> | 4.7           | 0.600            | 0.17        | 2.30           | 0.058     | 17.0                |
| <b>96</b> | 6.8           | 0.775            | 0.00        | 3.00           | 0.102     | 8.0                 |
| <b>97</b> | 7.0           | 0.500            | 0.25        | 2.00           | 0.070     | 3.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>98</b>  | 7.6           | 0.900            | 0.06        | 2.50           | 0.079     | 5.0                 |
| <b>99</b>  | 8.1           | 0.545            | 0.18        | 1.90           | 0.080     | 13.0                |
| <b>100</b> | 8.3           | 0.610            | 0.30        | 2.10           | 0.084     | 11.0                |
| <b>101</b> | 7.8           | 0.500            | 0.30        | 1.90           | 0.075     | 8.0                 |
| <b>103</b> | 8.1           | 0.575            | 0.22        | 2.10           | 0.077     | 12.0                |
| <b>104</b> | 7.2           | 0.490            | 0.24        | 2.20           | 0.070     | 5.0                 |
| <b>106</b> | 7.8           | 0.410            | 0.68        | 1.70           | 0.467     | 18.0                |
| <b>107</b> | 6.2           | 0.630            | 0.31        | 1.70           | 0.088     | 15.0                |
| <b>108</b> | 8.0           | 0.330            | 0.53        | 2.50           | 0.091     | 18.0                |
| <b>109</b> | 8.1           | 0.785            | 0.52        | 2.00           | 0.122     | 37.0                |
| <b>110</b> | 7.8           | 0.560            | 0.19        | 1.80           | 0.104     | 12.0                |
| <b>111</b> | 8.4           | 0.620            | 0.09        | 2.20           | 0.084     | 11.0                |
| <b>112</b> | 8.4           | 0.600            | 0.10        | 2.20           | 0.085     | 14.0                |
| <b>113</b> | 10.1          | 0.310            | 0.44        | 2.30           | 0.080     | 22.0                |
| <b>115</b> | 9.4           | 0.400            | 0.31        | 2.20           | 0.090     | 13.0                |
| <b>116</b> | 8.3           | 0.540            | 0.28        | 1.90           | 0.077     | 11.0                |
| <b>117</b> | 7.8           | 0.560            | 0.12        | 2.00           | 0.082     | 7.0                 |
| <b>118</b> | 8.8           | 0.550            | 0.04        | 2.20           | 0.119     | 14.0                |
| <b>119</b> | 7.0           | 0.690            | 0.08        | 1.80           | 0.097     | 22.0                |
| <b>120</b> | 7.3           | 1.070            | 0.09        | 1.70           | 0.178     | 10.0                |
| <b>122</b> | 7.3           | 0.695            | 0.00        | 2.50           | 0.075     | 3.0                 |
| <b>123</b> | 8.0           | 0.710            | 0.00        | 2.60           | 0.080     | 11.0                |
| <b>124</b> | 7.8           | 0.500            | 0.17        | 1.60           | 0.082     | 21.0                |
| <b>125</b> | 9.0           | 0.620            | 0.04        | 1.90           | 0.146     | 27.0                |
| <b>126</b> | 8.2           | 1.330            | 0.00        | 1.70           | 0.081     | 3.0                 |
| <b>127</b> | 8.1           | 1.330            | 0.00        | 1.80           | 0.082     | 3.0                 |
| <b>128</b> | 8.0           | 0.590            | 0.16        | 1.80           | 0.065     | 3.0                 |
| <b>129</b> | 6.1           | 0.380            | 0.15        | 1.80           | 0.072     | 6.0                 |
| <b>130</b> | 8.0           | 0.745            | 0.56        | 2.00           | 0.118     | 30.0                |
| <b>131</b> | 5.6           | 0.500            | 0.09        | 2.30           | 0.049     | 17.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>133</b> | 6.6           | 0.500            | 0.01        | 1.50           | 0.060     | 17.0                |
| <b>134</b> | 7.9           | 1.040            | 0.05        | 2.20           | 0.084     | 13.0                |
| <b>135</b> | 8.4           | 0.745            | 0.11        | 1.90           | 0.090     | 16.0                |
| <b>136</b> | 8.3           | 0.715            | 0.15        | 1.80           | 0.089     | 10.0                |
| <b>137</b> | 7.2           | 0.415            | 0.36        | 2.00           | 0.081     | 13.0                |
| <b>138</b> | 7.8           | 0.560            | 0.19        | 2.10           | 0.081     | 15.0                |
| <b>139</b> | 7.8           | 0.560            | 0.19        | 2.00           | 0.081     | 17.0                |
| <b>142</b> | 5.2           | 0.340            | 0.00        | 1.80           | 0.050     | 27.0                |
| <b>143</b> | 6.3           | 0.390            | 0.08        | 1.70           | 0.066     | 3.0                 |
| <b>145</b> | 8.1           | 0.670            | 0.55        | 1.80           | 0.117     | 32.0                |
| <b>146</b> | 5.8           | 0.680            | 0.02        | 1.80           | 0.087     | 21.0                |
| <b>147</b> | 7.6           | 0.490            | 0.26        | 1.60           | 0.236     | 10.0                |
| <b>148</b> | 6.9           | 0.490            | 0.10        | 2.30           | 0.074     | 12.0                |
| <b>149</b> | 8.2           | 0.400            | 0.44        | 2.80           | 0.089     | 11.0                |
| <b>150</b> | 7.3           | 0.330            | 0.47        | 2.10           | 0.077     | 5.0                 |
| <b>151</b> | 9.2           | 0.520            | 1.00        | 3.40           | 0.610     | 32.0                |
| <b>152</b> | 7.5           | 0.600            | 0.03        | 1.80           | 0.095     | 25.0                |
| <b>154</b> | 7.1           | 0.430            | 0.42        | 5.50           | 0.070     | 29.0                |
| <b>155</b> | 7.1           | 0.430            | 0.42        | 5.50           | 0.071     | 28.0                |
| <b>158</b> | 7.1           | 0.680            | 0.00        | 2.20           | 0.073     | 12.0                |
| <b>159</b> | 6.8           | 0.600            | 0.18        | 1.90           | 0.079     | 18.0                |
| <b>160</b> | 7.6           | 0.950            | 0.03        | 2.00           | 0.090     | 7.0                 |
| <b>161</b> | 7.6           | 0.680            | 0.02        | 1.30           | 0.072     | 9.0                 |
| <b>162</b> | 7.8           | 0.530            | 0.04        | 1.70           | 0.076     | 17.0                |
| <b>163</b> | 7.4           | 0.600            | 0.26        | 7.30           | 0.070     | 36.0                |
| <b>164</b> | 7.3           | 0.590            | 0.26        | 7.20           | 0.070     | 35.0                |
| <b>165</b> | 7.8           | 0.630            | 0.48        | 1.70           | 0.100     | 14.0                |
| <b>166</b> | 6.8           | 0.640            | 0.10        | 2.10           | 0.085     | 18.0                |
| <b>167</b> | 7.3           | 0.550            | 0.03        | 1.60           | 0.072     | 17.0                |
| <b>168</b> | 6.8           | 0.630            | 0.07        | 2.10           | 0.089     | 11.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>169</b> | 7.5           | 0.705            | 0.24        | 1.80           | 0.360     | 15.0                |
| <b>170</b> | 7.9           | 0.885            | 0.03        | 1.80           | 0.058     | 4.0                 |
| <b>171</b> | 8.0           | 0.420            | 0.17        | 2.00           | 0.073     | 6.0                 |
| <b>173</b> | 7.4           | 0.620            | 0.05        | 1.90           | 0.068     | 24.0                |
| <b>174</b> | 7.3           | 0.380            | 0.21        | 2.00           | 0.080     | 7.0                 |
| <b>175</b> | 6.9           | 0.500            | 0.04        | 1.50           | 0.085     | 19.0                |
| <b>177</b> | 7.5           | 0.520            | 0.42        | 2.30           | 0.087     | 8.0                 |
| <b>178</b> | 7.0           | 0.805            | 0.00        | 2.50           | 0.068     | 7.0                 |
| <b>179</b> | 8.8           | 0.610            | 0.14        | 2.40           | 0.067     | 10.0                |
| <b>181</b> | 8.9           | 0.610            | 0.49        | 2.00           | 0.270     | 23.0                |
| <b>182</b> | 7.2           | 0.730            | 0.02        | 2.50           | 0.076     | 16.0                |
| <b>183</b> | 6.8           | 0.610            | 0.20        | 1.80           | 0.077     | 11.0                |
| <b>184</b> | 6.7           | 0.620            | 0.21        | 1.90           | 0.079     | 8.0                 |
| <b>185</b> | 8.9           | 0.310            | 0.57        | 2.00           | 0.111     | 26.0                |
| <b>186</b> | 7.4           | 0.390            | 0.48        | 2.00           | 0.082     | 14.0                |
| <b>187</b> | 7.7           | 0.705            | 0.10        | 2.60           | 0.084     | 9.0                 |
| <b>188</b> | 7.9           | 0.500            | 0.33        | 2.00           | 0.084     | 15.0                |
| <b>189</b> | 7.9           | 0.490            | 0.32        | 1.90           | 0.082     | 17.0                |
| <b>190</b> | 8.2           | 0.500            | 0.35        | 2.90           | 0.077     | 21.0                |
| <b>191</b> | 6.4           | 0.370            | 0.25        | 1.90           | 0.074     | 21.0                |
| <b>192</b> | 6.8           | 0.630            | 0.12        | 3.80           | 0.099     | 16.0                |
| <b>193</b> | 7.6           | 0.550            | 0.21        | 2.20           | 0.071     | 7.0                 |
| <b>195</b> | 7.8           | 0.590            | 0.33        | 2.00           | 0.074     | 24.0                |
| <b>196</b> | 7.3           | 0.580            | 0.30        | 2.40           | 0.074     | 15.0                |
| <b>197</b> | 11.5          | 0.300            | 0.60        | 2.00           | 0.067     | 12.0                |
| <b>198</b> | 5.4           | 0.835            | 0.08        | 1.20           | 0.046     | 13.0                |
| <b>199</b> | 6.9           | 1.090            | 0.06        | 2.10           | 0.061     | 12.0                |
| <b>200</b> | 9.6           | 0.320            | 0.47        | 1.40           | 0.056     | 9.0                 |
| <b>201</b> | 8.8           | 0.370            | 0.48        | 2.10           | 0.097     | 39.0                |
| <b>202</b> | 6.8           | 0.500            | 0.11        | 1.50           | 0.075     | 16.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>203</b> | 7.0           | 0.420            | 0.35        | 1.60           | 0.088     | 16.0                |
| <b>204</b> | 7.0           | 0.430            | 0.36        | 1.60           | 0.089     | 14.0                |
| <b>205</b> | 12.8          | 0.300            | 0.74        | 2.60           | 0.095     | 9.0                 |
| <b>207</b> | 7.8           | 0.570            | 0.31        | 1.80           | 0.069     | 26.0                |
| <b>208</b> | 7.8           | 0.440            | 0.28        | 2.70           | 0.100     | 18.0                |
| <b>209</b> | 11.0          | 0.300            | 0.58        | 2.10           | 0.054     | 7.0                 |
| <b>210</b> | 9.7           | 0.530            | 0.60        | 2.00           | 0.039     | 5.0                 |
| <b>211</b> | 8.0           | 0.725            | 0.24        | 2.80           | 0.083     | 10.0                |
| <b>212</b> | 11.6          | 0.440            | 0.64        | 2.10           | 0.059     | 5.0                 |
| <b>213</b> | 8.2           | 0.570            | 0.26        | 2.20           | 0.060     | 28.0                |
| <b>214</b> | 7.8           | 0.735            | 0.08        | 2.40           | 0.092     | 10.0                |
| <b>215</b> | 7.0           | 0.490            | 0.49        | 5.60           | 0.060     | 26.0                |
| <b>216</b> | 8.7           | 0.625            | 0.16        | 2.00           | 0.101     | 13.0                |
| <b>217</b> | 8.1           | 0.725            | 0.22        | 2.20           | 0.072     | 11.0                |
| <b>218</b> | 7.5           | 0.490            | 0.19        | 1.90           | 0.076     | 10.0                |
| <b>219</b> | 7.8           | 0.530            | 0.33        | 2.40           | 0.080     | 24.0                |
| <b>220</b> | 7.8           | 0.340            | 0.37        | 2.00           | 0.082     | 24.0                |
| <b>221</b> | 7.4           | 0.530            | 0.26        | 2.00           | 0.101     | 16.0                |
| <b>222</b> | 6.8           | 0.610            | 0.04        | 1.50           | 0.057     | 5.0                 |
| <b>223</b> | 8.6           | 0.645            | 0.25        | 2.00           | 0.083     | 8.0                 |
| <b>224</b> | 8.4           | 0.635            | 0.36        | 2.00           | 0.089     | 15.0                |
| <b>225</b> | 7.7           | 0.430            | 0.25        | 2.60           | 0.073     | 29.0                |
| <b>226</b> | 8.9           | 0.590            | 0.50        | 2.00           | 0.337     | 27.0                |
| <b>227</b> | 9.0           | 0.820            | 0.14        | 2.60           | 0.089     | 9.0                 |
| <b>229</b> | 6.9           | 0.520            | 0.25        | 2.60           | 0.081     | 10.0                |
| <b>230</b> | 5.2           | 0.480            | 0.04        | 1.60           | 0.054     | 19.0                |
| <b>231</b> | 8.0           | 0.380            | 0.06        | 1.80           | 0.078     | 12.0                |
| <b>232</b> | 8.5           | 0.370            | 0.20        | 2.80           | 0.090     | 18.0                |
| <b>234</b> | 8.2           | 1.000            | 0.09        | 2.30           | 0.065     | 7.0                 |
| <b>235</b> | 7.2           | 0.630            | 0.00        | 1.90           | 0.097     | 14.0                |



|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>237</b> | 7.2           | 0.645            | 0.00        | 1.90           | 0.097     | 15.0                |
| <b>240</b> | 8.9           | 0.635            | 0.37        | 1.70           | 0.263     | 5.0                 |
| <b>241</b> | 12.0          | 0.380            | 0.56        | 2.10           | 0.093     | 6.0                 |
| <b>242</b> | 7.7           | 0.580            | 0.10        | 1.80           | 0.102     | 28.0                |
| <b>243</b> | 15.0          | 0.210            | 0.44        | 2.20           | 0.075     | 10.0                |
| <b>245</b> | 7.3           | 0.660            | 0.00        | 2.00           | 0.084     | 6.0                 |
| <b>246</b> | 7.1           | 0.680            | 0.07        | 1.90           | 0.075     | 16.0                |
| <b>247</b> | 8.2           | 0.600            | 0.17        | 2.30           | 0.072     | 11.0                |
| <b>248</b> | 7.7           | 0.530            | 0.06        | 1.70           | 0.074     | 9.0                 |
| <b>250</b> | 10.8          | 0.320            | 0.44        | 1.60           | 0.063     | 16.0                |
| <b>251</b> | 7.1           | 0.600            | 0.00        | 1.80           | 0.074     | 16.0                |
| <b>252</b> | 11.1          | 0.350            | 0.48        | 3.10           | 0.090     | 5.0                 |
| <b>253</b> | 7.7           | 0.775            | 0.42        | 1.90           | 0.092     | 8.0                 |
| <b>255</b> | 8.0           | 0.570            | 0.23        | 3.20           | 0.073     | 17.0                |
| <b>256</b> | 9.4           | 0.340            | 0.37        | 2.20           | 0.075     | 5.0                 |
| <b>257</b> | 6.6           | 0.695            | 0.00        | 2.10           | 0.075     | 12.0                |
| <b>258</b> | 7.7           | 0.410            | 0.76        | 1.80           | 0.611     | 8.0                 |
| <b>259</b> | 10.0          | 0.310            | 0.47        | 2.60           | 0.085     | 14.0                |
| <b>260</b> | 7.9           | 0.330            | 0.23        | 1.70           | 0.077     | 18.0                |
| <b>261</b> | 7.0           | 0.975            | 0.04        | 2.00           | 0.087     | 12.0                |
| <b>262</b> | 8.0           | 0.520            | 0.03        | 1.70           | 0.070     | 10.0                |
| <b>263</b> | 7.9           | 0.370            | 0.23        | 1.80           | 0.077     | 23.0                |
| <b>264</b> | 12.5          | 0.560            | 0.49        | 2.40           | 0.064     | 5.0                 |
| <b>265</b> | 11.8          | 0.260            | 0.52        | 1.80           | 0.071     | 6.0                 |
| <b>266</b> | 8.1           | 0.870            | 0.00        | 3.30           | 0.096     | 26.0                |
| <b>267</b> | 7.9           | 0.350            | 0.46        | 3.60           | 0.078     | 15.0                |
| <b>268</b> | 6.9           | 0.540            | 0.04        | 3.00           | 0.077     | 7.0                 |
| <b>269</b> | 11.5          | 0.180            | 0.51        | 4.00           | 0.104     | 4.0                 |
| <b>270</b> | 7.9           | 0.545            | 0.06        | 4.00           | 0.087     | 27.0                |
| <b>272</b> | 10.9          | 0.370            | 0.58        | 4.00           | 0.071     | 17.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>273</b> | 8.4           | 0.715            | 0.20        | 2.40           | 0.076     | 10.0                |
| <b>274</b> | 7.5           | 0.650            | 0.18        | 7.00           | 0.088     | 27.0                |
| <b>278</b> | 10.3          | 0.320            | 0.45        | 6.40           | 0.073     | 5.0                 |
| <b>279</b> | 8.9           | 0.400            | 0.32        | 5.60           | 0.087     | 10.0                |
| <b>280</b> | 11.4          | 0.260            | 0.44        | 3.60           | 0.071     | 6.0                 |
| <b>281</b> | 7.7           | 0.270            | 0.68        | 3.50           | 0.358     | 5.0                 |
| <b>282</b> | 7.6           | 0.520            | 0.12        | 3.00           | 0.067     | 12.0                |
| <b>284</b> | 9.9           | 0.590            | 0.07        | 3.40           | 0.102     | 32.0                |
| <b>286</b> | 12.0          | 0.450            | 0.55        | 2.00           | 0.073     | 25.0                |
| <b>287</b> | 7.5           | 0.400            | 0.12        | 3.00           | 0.092     | 29.0                |
| <b>288</b> | 8.7           | 0.520            | 0.09        | 2.50           | 0.091     | 20.0                |
| <b>289</b> | 11.6          | 0.420            | 0.53        | 3.30           | 0.105     | 33.0                |
| <b>291</b> | 11.0          | 0.200            | 0.48        | 2.00           | 0.343     | 6.0                 |
| <b>292</b> | 10.4          | 0.550            | 0.23        | 2.70           | 0.091     | 18.0                |
| <b>293</b> | 6.9           | 0.360            | 0.25        | 2.40           | 0.098     | 5.0                 |
| <b>294</b> | 13.3          | 0.340            | 0.52        | 3.20           | 0.094     | 17.0                |
| <b>295</b> | 10.8          | 0.500            | 0.46        | 2.50           | 0.073     | 5.0                 |
| <b>296</b> | 10.6          | 0.830            | 0.37        | 2.60           | 0.086     | 26.0                |
| <b>297</b> | 7.1           | 0.630            | 0.06        | 2.00           | 0.083     | 8.0                 |
| <b>298</b> | 7.2           | 0.650            | 0.02        | 2.30           | 0.094     | 5.0                 |
| <b>299</b> | 6.9           | 0.670            | 0.06        | 2.10           | 0.080     | 8.0                 |
| <b>300</b> | 7.5           | 0.530            | 0.06        | 2.60           | 0.086     | 20.0                |
| <b>301</b> | 11.1          | 0.180            | 0.48        | 1.50           | 0.068     | 7.0                 |
| <b>302</b> | 8.3           | 0.705            | 0.12        | 2.60           | 0.092     | 12.0                |
| <b>303</b> | 7.4           | 0.670            | 0.12        | 1.60           | 0.186     | 5.0                 |
| <b>304</b> | 8.4           | 0.650            | 0.60        | 2.10           | 0.112     | 12.0                |
| <b>305</b> | 10.3          | 0.530            | 0.48        | 2.50           | 0.063     | 6.0                 |
| <b>306</b> | 7.6           | 0.620            | 0.32        | 2.20           | 0.082     | 7.0                 |
| <b>307</b> | 10.3          | 0.410            | 0.42        | 2.40           | 0.213     | 6.0                 |
| <b>308</b> | 10.3          | 0.430            | 0.44        | 2.40           | 0.214     | 5.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>309</b> | 7.4           | 0.290            | 0.38        | 1.70           | 0.062     | 9.0                 |
| <b>311</b> | 7.9           | 0.530            | 0.24        | 2.00           | 0.072     | 15.0                |
| <b>312</b> | 9.0           | 0.460            | 0.31        | 2.80           | 0.093     | 19.0                |
| <b>313</b> | 8.6           | 0.470            | 0.30        | 3.00           | 0.076     | 30.0                |
| <b>314</b> | 7.4           | 0.360            | 0.29        | 2.60           | 0.087     | 26.0                |
| <b>315</b> | 7.1           | 0.350            | 0.29        | 2.50           | 0.096     | 20.0                |
| <b>316</b> | 9.6           | 0.560            | 0.23        | 3.40           | 0.102     | 37.0                |
| <b>317</b> | 9.6           | 0.770            | 0.12        | 2.90           | 0.082     | 30.0                |
| <b>318</b> | 9.8           | 0.660            | 0.39        | 3.20           | 0.083     | 21.0                |
| <b>321</b> | 9.3           | 0.610            | 0.26        | 3.40           | 0.090     | 25.0                |
| <b>322</b> | 7.8           | 0.620            | 0.05        | 2.30           | 0.079     | 6.0                 |
| <b>323</b> | 10.3          | 0.590            | 0.42        | 2.80           | 0.090     | 35.0                |
| <b>324</b> | 10.0          | 0.490            | 0.20        | 11.00          | 0.071     | 13.0                |
| <b>326</b> | 11.6          | 0.530            | 0.66        | 3.65           | 0.121     | 6.0                 |
| <b>327</b> | 10.3          | 0.440            | 0.50        | 4.50           | 0.107     | 5.0                 |
| <b>328</b> | 13.4          | 0.270            | 0.62        | 2.60           | 0.082     | 6.0                 |
| <b>329</b> | 10.7          | 0.460            | 0.39        | 2.00           | 0.061     | 7.0                 |
| <b>330</b> | 10.2          | 0.360            | 0.64        | 2.90           | 0.122     | 10.0                |
| <b>332</b> | 8.0           | 0.580            | 0.28        | 3.20           | 0.066     | 21.0                |
| <b>333</b> | 8.4           | 0.560            | 0.08        | 2.10           | 0.105     | 16.0                |
| <b>334</b> | 7.9           | 0.650            | 0.01        | 2.50           | 0.078     | 17.0                |
| <b>335</b> | 11.9          | 0.695            | 0.53        | 3.40           | 0.128     | 7.0                 |
| <b>336</b> | 8.9           | 0.430            | 0.45        | 1.90           | 0.052     | 6.0                 |
| <b>337</b> | 7.8           | 0.430            | 0.32        | 2.80           | 0.080     | 29.0                |
| <b>338</b> | 12.4          | 0.490            | 0.58        | 3.00           | 0.103     | 28.0                |
| <b>339</b> | 12.5          | 0.280            | 0.54        | 2.30           | 0.082     | 12.0                |
| <b>340</b> | 12.2          | 0.340            | 0.50        | 2.40           | 0.066     | 10.0                |
| <b>341</b> | 10.6          | 0.420            | 0.48        | 2.70           | 0.065     | 5.0                 |
| <b>342</b> | 10.9          | 0.390            | 0.47        | 1.80           | 0.118     | 6.0                 |
| <b>344</b> | 11.9          | 0.570            | 0.50        | 2.60           | 0.082     | 6.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>345</b> | 7.0           | 0.685            | 0.00        | 1.90           | 0.067     | 40.0                |
| <b>346</b> | 6.6           | 0.815            | 0.02        | 2.70           | 0.072     | 17.0                |
| <b>347</b> | 13.8          | 0.490            | 0.67        | 3.00           | 0.093     | 6.0                 |
| <b>348</b> | 9.6           | 0.560            | 0.31        | 2.80           | 0.089     | 15.0                |
| <b>349</b> | 9.1           | 0.785            | 0.00        | 2.60           | 0.093     | 11.0                |
| <b>350</b> | 10.7          | 0.670            | 0.22        | 2.70           | 0.107     | 17.0                |
| <b>351</b> | 9.1           | 0.795            | 0.00        | 2.60           | 0.096     | 11.0                |
| <b>352</b> | 7.7           | 0.665            | 0.00        | 2.40           | 0.090     | 8.0                 |
| <b>353</b> | 13.5          | 0.530            | 0.79        | 4.80           | 0.120     | 23.0                |
| <b>354</b> | 6.1           | 0.210            | 0.40        | 1.40           | 0.066     | 40.5                |
| <b>355</b> | 6.7           | 0.750            | 0.01        | 2.40           | 0.078     | 17.0                |
| <b>356</b> | 11.5          | 0.410            | 0.52        | 3.00           | 0.080     | 29.0                |
| <b>357</b> | 10.5          | 0.420            | 0.66        | 2.95           | 0.116     | 12.0                |
| <b>358</b> | 11.9          | 0.430            | 0.66        | 3.10           | 0.109     | 10.0                |
| <b>359</b> | 12.6          | 0.380            | 0.66        | 2.60           | 0.088     | 10.0                |
| <b>360</b> | 8.2           | 0.700            | 0.23        | 2.00           | 0.099     | 14.0                |
| <b>361</b> | 8.6           | 0.450            | 0.31        | 2.60           | 0.086     | 21.0                |
| <b>362</b> | 11.9          | 0.580            | 0.66        | 2.50           | 0.072     | 6.0                 |
| <b>363</b> | 12.5          | 0.460            | 0.63        | 2.00           | 0.071     | 6.0                 |
| <b>364</b> | 12.8          | 0.615            | 0.66        | 5.80           | 0.083     | 7.0                 |
| <b>365</b> | 10.0          | 0.420            | 0.50        | 3.40           | 0.107     | 7.0                 |
| <b>367</b> | 10.4          | 0.575            | 0.61        | 2.60           | 0.076     | 11.0                |
| <b>368</b> | 10.3          | 0.340            | 0.52        | 2.80           | 0.159     | 15.0                |
| <b>369</b> | 9.4           | 0.270            | 0.53        | 2.40           | 0.074     | 6.0                 |
| <b>370</b> | 6.9           | 0.765            | 0.02        | 2.30           | 0.063     | 35.0                |
| <b>371</b> | 7.9           | 0.240            | 0.40        | 1.60           | 0.056     | 11.0                |
| <b>372</b> | 9.1           | 0.280            | 0.48        | 1.80           | 0.067     | 26.0                |
| <b>373</b> | 7.4           | 0.550            | 0.22        | 2.20           | 0.106     | 12.0                |
| <b>374</b> | 14.0          | 0.410            | 0.63        | 3.80           | 0.089     | 6.0                 |
| <b>375</b> | 11.5          | 0.540            | 0.71        | 4.40           | 0.124     | 6.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>376</b> | 11.5          | 0.450            | 0.50        | 3.00           | 0.078     | 19.0                |
| <b>378</b> | 11.4          | 0.625            | 0.66        | 6.20           | 0.088     | 6.0                 |
| <b>379</b> | 8.3           | 0.420            | 0.38        | 2.50           | 0.094     | 24.0                |
| <b>380</b> | 8.3           | 0.260            | 0.42        | 2.00           | 0.080     | 11.0                |
| <b>381</b> | 13.7          | 0.415            | 0.68        | 2.90           | 0.085     | 17.0                |
| <b>384</b> | 7.7           | 0.510            | 0.28        | 2.10           | 0.087     | 23.0                |
| <b>385</b> | 7.4           | 0.630            | 0.07        | 2.40           | 0.090     | 11.0                |
| <b>386</b> | 7.8           | 0.540            | 0.26        | 2.00           | 0.088     | 23.0                |
| <b>387</b> | 8.3           | 0.660            | 0.15        | 1.90           | 0.079     | 17.0                |
| <b>388</b> | 7.8           | 0.460            | 0.26        | 1.90           | 0.088     | 23.0                |
| <b>389</b> | 9.6           | 0.380            | 0.31        | 2.50           | 0.096     | 16.0                |
| <b>390</b> | 5.6           | 0.850            | 0.05        | 1.40           | 0.045     | 12.0                |
| <b>392</b> | 9.5           | 0.370            | 0.52        | 2.00           | 0.082     | 6.0                 |
| <b>393</b> | 8.4           | 0.665            | 0.61        | 2.00           | 0.112     | 13.0                |
| <b>394</b> | 12.7          | 0.600            | 0.65        | 2.30           | 0.063     | 6.0                 |
| <b>395</b> | 12.0          | 0.370            | 0.76        | 4.20           | 0.066     | 7.0                 |
| <b>396</b> | 6.6           | 0.735            | 0.02        | 7.90           | 0.122     | 68.0                |
| <b>397</b> | 11.5          | 0.590            | 0.59        | 2.60           | 0.087     | 13.0                |
| <b>399</b> | 8.7           | 0.765            | 0.22        | 2.30           | 0.064     | 9.0                 |
| <b>401</b> | 7.7           | 0.260            | 0.30        | 1.70           | 0.059     | 20.0                |
| <b>402</b> | 12.2          | 0.480            | 0.54        | 2.60           | 0.085     | 19.0                |
| <b>403</b> | 11.4          | 0.600            | 0.49        | 2.70           | 0.085     | 10.0                |
| <b>404</b> | 7.7           | 0.690            | 0.05        | 2.70           | 0.075     | 15.0                |
| <b>405</b> | 8.7           | 0.310            | 0.46        | 1.40           | 0.059     | 11.0                |
| <b>406</b> | 9.8           | 0.440            | 0.47        | 2.50           | 0.063     | 9.0                 |
| <b>407</b> | 12.0          | 0.390            | 0.66        | 3.00           | 0.093     | 12.0                |
| <b>408</b> | 10.4          | 0.340            | 0.58        | 3.70           | 0.174     | 6.0                 |
| <b>409</b> | 12.5          | 0.460            | 0.49        | 4.50           | 0.070     | 26.0                |
| <b>410</b> | 9.0           | 0.430            | 0.34        | 2.50           | 0.080     | 26.0                |
| <b>411</b> | 9.1           | 0.450            | 0.35        | 2.40           | 0.080     | 23.0                |

|     | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-----|---------------|------------------|-------------|----------------|-----------|---------------------|
| 412 | 7.1           | 0.735            | 0.16        | 1.90           | 0.100     | 15.0                |
| 413 | 9.9           | 0.400            | 0.53        | 6.70           | 0.097     | 6.0                 |
| 414 | 8.8           | 0.520            | 0.34        | 2.70           | 0.087     | 24.0                |
| 415 | 8.6           | 0.725            | 0.24        | 6.60           | 0.117     | 31.0                |
| 416 | 10.6          | 0.480            | 0.64        | 2.20           | 0.111     | 6.0                 |
| 417 | 7.0           | 0.580            | 0.12        | 1.90           | 0.091     | 34.0                |
| 418 | 11.9          | 0.380            | 0.51        | 2.00           | 0.121     | 7.0                 |
| 419 | 6.8           | 0.770            | 0.00        | 1.80           | 0.066     | 34.0                |
| 420 | 9.5           | 0.560            | 0.33        | 2.40           | 0.089     | 35.0                |
| 421 | 6.6           | 0.840            | 0.03        | 2.30           | 0.059     | 32.0                |
| 422 | 7.7           | 0.960            | 0.20        | 2.00           | 0.047     | 15.0                |
| 423 | 10.5          | 0.240            | 0.47        | 2.10           | 0.066     | 6.0                 |
| 426 | 6.4           | 0.670            | 0.08        | 2.10           | 0.045     | 19.0                |
| 427 | 9.5           | 0.780            | 0.22        | 1.90           | 0.077     | 6.0                 |
| 428 | 9.1           | 0.520            | 0.33        | 1.30           | 0.070     | 9.0                 |
| 429 | 12.8          | 0.840            | 0.63        | 2.40           | 0.088     | 13.0                |
| 431 | 7.8           | 0.550            | 0.35        | 2.20           | 0.074     | 21.0                |
| 432 | 11.9          | 0.370            | 0.69        | 2.30           | 0.078     | 12.0                |
| 433 | 12.3          | 0.390            | 0.63        | 2.30           | 0.091     | 6.0                 |
| 434 | 10.4          | 0.410            | 0.55        | 3.20           | 0.076     | 22.0                |
| 436 | 8.0           | 0.670            | 0.30        | 2.00           | 0.060     | 38.0                |
| 437 | 11.1          | 0.450            | 0.73        | 3.20           | 0.066     | 6.0                 |
| 439 | 7.0           | 0.620            | 0.18        | 1.50           | 0.062     | 7.0                 |
| 440 | 12.6          | 0.310            | 0.72        | 2.20           | 0.072     | 6.0                 |
| 441 | 11.9          | 0.400            | 0.65        | 2.15           | 0.068     | 7.0                 |
| 442 | 15.6          | 0.685            | 0.76        | 3.70           | 0.100     | 6.0                 |
| 443 | 10.0          | 0.440            | 0.49        | 2.70           | 0.077     | 11.0                |
| 444 | 5.3           | 0.570            | 0.01        | 1.70           | 0.054     | 5.0                 |
| 445 | 9.5           | 0.735            | 0.10        | 2.10           | 0.079     | 6.0                 |
| 446 | 12.5          | 0.380            | 0.60        | 2.60           | 0.081     | 31.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>447</b> | 9.3           | 0.480            | 0.29        | 2.10           | 0.127     | 6.0                 |
| <b>448</b> | 8.6           | 0.530            | 0.22        | 2.00           | 0.100     | 7.0                 |
| <b>449</b> | 11.9          | 0.390            | 0.69        | 2.80           | 0.095     | 17.0                |
| <b>451</b> | 8.4           | 0.370            | 0.53        | 1.80           | 0.413     | 9.0                 |
| <b>452</b> | 6.8           | 0.560            | 0.03        | 1.70           | 0.084     | 18.0                |
| <b>453</b> | 10.4          | 0.330            | 0.63        | 2.80           | 0.084     | 5.0                 |
| <b>454</b> | 7.0           | 0.230            | 0.40        | 1.60           | 0.063     | 21.0                |
| <b>455</b> | 11.3          | 0.620            | 0.67        | 5.20           | 0.086     | 6.0                 |
| <b>456</b> | 8.9           | 0.590            | 0.39        | 2.30           | 0.095     | 5.0                 |
| <b>457</b> | 9.2           | 0.630            | 0.21        | 2.70           | 0.097     | 29.0                |
| <b>459</b> | 11.6          | 0.580            | 0.66        | 2.20           | 0.074     | 10.0                |
| <b>460</b> | 9.2           | 0.430            | 0.52        | 2.30           | 0.083     | 14.0                |
| <b>461</b> | 8.3           | 0.615            | 0.22        | 2.60           | 0.087     | 6.0                 |
| <b>462</b> | 11.0          | 0.260            | 0.68        | 2.55           | 0.085     | 10.0                |
| <b>463</b> | 8.1           | 0.660            | 0.70        | 2.20           | 0.098     | 25.0                |
| <b>464</b> | 11.5          | 0.315            | 0.54        | 2.10           | 0.084     | 5.0                 |
| <b>465</b> | 10.0          | 0.290            | 0.40        | 2.90           | 0.098     | 10.0                |
| <b>466</b> | 10.3          | 0.500            | 0.42        | 2.00           | 0.069     | 21.0                |
| <b>467</b> | 8.8           | 0.460            | 0.45        | 2.60           | 0.065     | 7.0                 |
| <b>468</b> | 11.4          | 0.360            | 0.69        | 2.10           | 0.090     | 6.0                 |
| <b>469</b> | 8.7           | 0.820            | 0.02        | 1.20           | 0.070     | 36.0                |
| <b>470</b> | 13.0          | 0.320            | 0.65        | 2.60           | 0.093     | 15.0                |
| <b>471</b> | 9.6           | 0.540            | 0.42        | 2.40           | 0.081     | 25.0                |
| <b>472</b> | 12.5          | 0.370            | 0.55        | 2.60           | 0.083     | 25.0                |
| <b>473</b> | 9.9           | 0.350            | 0.55        | 2.10           | 0.062     | 5.0                 |
| <b>474</b> | 10.5          | 0.280            | 0.51        | 1.70           | 0.080     | 10.0                |
| <b>475</b> | 9.6           | 0.680            | 0.24        | 2.20           | 0.087     | 5.0                 |
| <b>476</b> | 9.3           | 0.270            | 0.41        | 2.00           | 0.091     | 6.0                 |
| <b>477</b> | 10.4          | 0.240            | 0.49        | 1.80           | 0.075     | 6.0                 |
| <b>479</b> | 9.4           | 0.685            | 0.11        | 2.70           | 0.077     | 6.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>480</b> | 10.6          | 0.280            | 0.39        | 15.50          | 0.069     | 6.0                 |
| <b>481</b> | 9.4           | 0.300            | 0.56        | 2.80           | 0.080     | 6.0                 |
| <b>482</b> | 10.6          | 0.360            | 0.59        | 2.20           | 0.152     | 6.0                 |
| <b>483</b> | 10.6          | 0.360            | 0.60        | 2.20           | 0.152     | 7.0                 |
| <b>484</b> | 10.6          | 0.440            | 0.68        | 4.10           | 0.114     | 6.0                 |
| <b>485</b> | 10.2          | 0.670            | 0.39        | 1.90           | 0.054     | 6.0                 |
| <b>487</b> | 10.2          | 0.645            | 0.36        | 1.80           | 0.053     | 5.0                 |
| <b>488</b> | 11.6          | 0.320            | 0.55        | 2.80           | 0.081     | 35.0                |
| <b>489</b> | 9.3           | 0.390            | 0.40        | 2.60           | 0.073     | 10.0                |
| <b>490</b> | 9.3           | 0.775            | 0.27        | 2.80           | 0.078     | 24.0                |
| <b>491</b> | 9.2           | 0.410            | 0.50        | 2.50           | 0.055     | 12.0                |
| <b>492</b> | 8.9           | 0.400            | 0.51        | 2.60           | 0.052     | 13.0                |
| <b>493</b> | 8.7           | 0.690            | 0.31        | 3.00           | 0.086     | 23.0                |
| <b>494</b> | 6.5           | 0.390            | 0.23        | 8.30           | 0.051     | 28.0                |
| <b>495</b> | 10.7          | 0.350            | 0.53        | 2.60           | 0.070     | 5.0                 |
| <b>496</b> | 7.8           | 0.520            | 0.25        | 1.90           | 0.081     | 14.0                |
| <b>497</b> | 7.2           | 0.340            | 0.32        | 2.50           | 0.090     | 43.0                |
| <b>501</b> | 10.4          | 0.440            | 0.73        | 6.55           | 0.074     | 38.0                |
| <b>503</b> | 10.5          | 0.260            | 0.47        | 1.90           | 0.078     | 6.0                 |
| <b>504</b> | 10.5          | 0.240            | 0.42        | 1.80           | 0.077     | 6.0                 |
| <b>505</b> | 10.2          | 0.490            | 0.63        | 2.90           | 0.072     | 10.0                |
| <b>506</b> | 10.4          | 0.240            | 0.46        | 1.80           | 0.075     | 6.0                 |
| <b>507</b> | 11.2          | 0.670            | 0.55        | 2.30           | 0.084     | 6.0                 |
| <b>508</b> | 10.0          | 0.590            | 0.31        | 2.20           | 0.090     | 26.0                |
| <b>509</b> | 13.3          | 0.290            | 0.75        | 2.80           | 0.084     | 23.0                |
| <b>510</b> | 12.4          | 0.420            | 0.49        | 4.60           | 0.073     | 19.0                |
| <b>512</b> | 10.7          | 0.400            | 0.48        | 2.10           | 0.125     | 15.0                |
| <b>513</b> | 10.5          | 0.510            | 0.64        | 2.40           | 0.107     | 6.0                 |
| <b>515</b> | 8.5           | 0.655            | 0.49        | 6.10           | 0.122     | 34.0                |
| <b>516</b> | 12.5          | 0.600            | 0.49        | 4.30           | 0.100     | 5.0                 |



|     | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-----|---------------|------------------|-------------|----------------|-----------|---------------------|
| 517 | 10.4          | 0.610            | 0.49        | 2.10           | 0.200     | 5.0                 |
| 518 | 10.9          | 0.210            | 0.49        | 2.80           | 0.088     | 11.0                |
| 519 | 7.3           | 0.365            | 0.49        | 2.50           | 0.088     | 39.0                |
| 520 | 9.8           | 0.250            | 0.49        | 2.70           | 0.088     | 15.0                |
| 521 | 7.6           | 0.410            | 0.49        | 2.00           | 0.088     | 16.0                |
| 522 | 8.2           | 0.390            | 0.49        | 2.30           | 0.099     | 47.0                |
| 523 | 9.3           | 0.400            | 0.49        | 2.50           | 0.085     | 38.0                |
| 524 | 9.2           | 0.430            | 0.49        | 2.40           | 0.086     | 23.0                |
| 525 | 10.4          | 0.640            | 0.24        | 2.80           | 0.105     | 29.0                |
| 527 | 7.0           | 0.380            | 0.49        | 2.50           | 0.097     | 33.0                |
| 528 | 8.2           | 0.420            | 0.49        | 2.60           | 0.084     | 32.0                |
| 529 | 9.9           | 0.630            | 0.24        | 2.40           | 0.077     | 6.0                 |
| 530 | 9.1           | 0.220            | 0.24        | 2.10           | 0.078     | 1.0                 |
| 531 | 11.9          | 0.380            | 0.49        | 2.70           | 0.098     | 12.0                |
| 533 | 10.3          | 0.270            | 0.24        | 2.10           | 0.072     | 15.0                |
| 534 | 10.0          | 0.480            | 0.24        | 2.70           | 0.102     | 13.0                |
| 537 | 8.1           | 0.825            | 0.24        | 2.10           | 0.084     | 5.0                 |
| 538 | 12.9          | 0.350            | 0.49        | 5.80           | 0.066     | 5.0                 |
| 539 | 11.2          | 0.500            | 0.74        | 5.15           | 0.100     | 5.0                 |
| 540 | 9.2           | 0.590            | 0.24        | 3.30           | 0.101     | 20.0                |
| 541 | 9.5           | 0.460            | 0.49        | 6.30           | 0.064     | 5.0                 |
| 542 | 9.3           | 0.715            | 0.24        | 2.10           | 0.070     | 5.0                 |
| 543 | 11.2          | 0.660            | 0.24        | 2.50           | 0.085     | 16.0                |
| 544 | 14.3          | 0.310            | 0.74        | 1.80           | 0.075     | 6.0                 |
| 545 | 9.1           | 0.470            | 0.49        | 2.60           | 0.094     | 38.0                |
| 546 | 7.5           | 0.550            | 0.24        | 2.00           | 0.078     | 10.0                |
| 547 | 10.6          | 0.310            | 0.49        | 2.50           | 0.067     | 6.0                 |
| 548 | 12.4          | 0.350            | 0.49        | 2.60           | 0.079     | 27.0                |
| 549 | 9.0           | 0.530            | 0.49        | 1.90           | 0.171     | 6.0                 |
| 550 | 6.8           | 0.510            | 0.01        | 2.10           | 0.074     | 9.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>551</b> | 9.4           | 0.430            | 0.24        | 2.80           | 0.092     | 14.0                |
| <b>552</b> | 9.5           | 0.460            | 0.24        | 2.70           | 0.092     | 14.0                |
| <b>553</b> | 5.0           | 1.040            | 0.24        | 1.60           | 0.050     | 32.0                |
| <b>554</b> | 15.5          | 0.645            | 0.49        | 4.20           | 0.095     | 10.0                |
| <b>556</b> | 10.9          | 0.530            | 0.49        | 4.60           | 0.118     | 10.0                |
| <b>557</b> | 15.6          | 0.645            | 0.49        | 4.20           | 0.095     | 10.0                |
| <b>559</b> | 13.0          | 0.470            | 0.49        | 4.30           | 0.085     | 6.0                 |
| <b>560</b> | 12.7          | 0.600            | 0.49        | 2.80           | 0.075     | 5.0                 |
| <b>561</b> | 9.0           | 0.440            | 0.49        | 2.40           | 0.078     | 26.0                |
| <b>562</b> | 9.0           | 0.540            | 0.49        | 2.90           | 0.094     | 41.0                |
| <b>563</b> | 7.6           | 0.290            | 0.49        | 2.70           | 0.092     | 25.0                |
| <b>566</b> | 8.7           | 0.700            | 0.24        | 2.50           | 0.226     | 5.0                 |
| <b>568</b> | 9.8           | 0.500            | 0.49        | 2.60           | 0.250     | 5.0                 |
| <b>569</b> | 6.2           | 0.360            | 0.24        | 2.20           | 0.095     | 19.0                |
| <b>570</b> | 11.5          | 0.350            | 0.49        | 3.30           | 0.070     | 10.0                |
| <b>572</b> | 10.2          | 0.240            | 0.49        | 2.40           | 0.075     | 10.0                |
| <b>573</b> | 10.5          | 0.590            | 0.49        | 2.10           | 0.070     | 14.0                |
| <b>574</b> | 10.6          | 0.340            | 0.49        | 3.20           | 0.078     | 20.0                |
| <b>575</b> | 12.3          | 0.270            | 0.49        | 3.10           | 0.079     | 28.0                |
| <b>576</b> | 9.9           | 0.500            | 0.24        | 2.30           | 0.103     | 6.0                 |
| <b>577</b> | 8.8           | 0.440            | 0.49        | 2.80           | 0.083     | 18.0                |
| <b>578</b> | 8.8           | 0.470            | 0.49        | 2.90           | 0.085     | 17.0                |
| <b>579</b> | 10.6          | 0.310            | 0.49        | 2.20           | 0.063     | 18.0                |
| <b>580</b> | 12.3          | 0.500            | 0.49        | 2.20           | 0.089     | 5.0                 |
| <b>582</b> | 11.7          | 0.490            | 0.49        | 2.20           | 0.083     | 5.0                 |
| <b>583</b> | 12.0          | 0.280            | 0.49        | 1.90           | 0.074     | 10.0                |
| <b>584</b> | 11.8          | 0.330            | 0.49        | 3.40           | 0.093     | 54.0                |
| <b>585</b> | 7.6           | 0.510            | 0.24        | 2.40           | 0.091     | 8.0                 |
| <b>586</b> | 11.1          | 0.310            | 0.49        | 2.70           | 0.094     | 16.0                |
| <b>587</b> | 7.3           | 0.730            | 0.24        | 1.90           | 0.108     | 18.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>588</b> | 5.0           | 0.420            | 0.24        | 2.00           | 0.060     | 19.0                |
| <b>589</b> | 10.2          | 0.290            | 0.49        | 2.60           | 0.059     | 5.0                 |
| <b>590</b> | 9.0           | 0.450            | 0.49        | 2.60           | 0.084     | 21.0                |
| <b>591</b> | 6.6           | 0.390            | 0.49        | 1.70           | 0.070     | 23.0                |
| <b>593</b> | 9.9           | 0.490            | 0.58        | 3.50           | 0.094     | 9.0                 |
| <b>594</b> | 7.9           | 0.720            | 0.17        | 2.60           | 0.096     | 20.0                |
| <b>595</b> | 8.9           | 0.595            | 0.41        | 7.90           | 0.086     | 30.0                |
| <b>596</b> | 12.4          | 0.400            | 0.51        | 2.00           | 0.059     | 6.0                 |
| <b>597</b> | 11.9          | 0.580            | 0.58        | 1.90           | 0.071     | 5.0                 |
| <b>598</b> | 8.5           | 0.585            | 0.18        | 2.10           | 0.078     | 5.0                 |
| <b>599</b> | 12.7          | 0.590            | 0.45        | 2.30           | 0.082     | 11.0                |
| <b>600</b> | 8.2           | 0.915            | 0.27        | 2.10           | 0.088     | 7.0                 |
| <b>601</b> | 13.2          | 0.460            | 0.52        | 2.20           | 0.071     | 12.0                |
| <b>602</b> | 7.7           | 0.835            | 0.00        | 2.60           | 0.081     | 6.0                 |
| <b>604</b> | 8.3           | 0.580            | 0.13        | 2.90           | 0.096     | 14.0                |
| <b>605</b> | 8.3           | 0.600            | 0.13        | 2.60           | 0.085     | 6.0                 |
| <b>606</b> | 9.4           | 0.410            | 0.48        | 4.60           | 0.072     | 10.0                |
| <b>607</b> | 8.8           | 0.480            | 0.41        | 3.30           | 0.092     | 26.0                |
| <b>608</b> | 10.1          | 0.650            | 0.37        | 5.10           | 0.110     | 11.0                |
| <b>609</b> | 6.3           | 0.360            | 0.19        | 3.20           | 0.075     | 15.0                |
| <b>610</b> | 8.8           | 0.240            | 0.54        | 2.50           | 0.083     | 25.0                |
| <b>611</b> | 13.2          | 0.380            | 0.55        | 2.70           | 0.081     | 5.0                 |
| <b>612</b> | 7.5           | 0.640            | 0.00        | 2.40           | 0.077     | 18.0                |
| <b>613</b> | 8.2           | 0.390            | 0.38        | 1.50           | 0.058     | 10.0                |
| <b>614</b> | 9.2           | 0.755            | 0.18        | 2.20           | 0.148     | 10.0                |
| <b>615</b> | 9.6           | 0.600            | 0.50        | 2.30           | 0.079     | 28.0                |
| <b>617</b> | 11.5          | 0.310            | 0.51        | 2.20           | 0.079     | 14.0                |
| <b>618</b> | 11.4          | 0.460            | 0.50        | 2.70           | 0.122     | 4.0                 |
| <b>619</b> | 11.3          | 0.370            | 0.41        | 2.30           | 0.088     | 6.0                 |
| <b>620</b> | 8.3           | 0.540            | 0.24        | 3.40           | 0.076     | 16.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>621</b> | 8.2           | 0.560            | 0.23        | 3.40           | 0.078     | 14.0                |
| <b>622</b> | 10.0          | 0.580            | 0.22        | 1.90           | 0.080     | 9.0                 |
| <b>623</b> | 7.9           | 0.510            | 0.25        | 2.90           | 0.077     | 21.0                |
| <b>624</b> | 6.8           | 0.690            | 0.00        | 5.60           | 0.124     | 21.0                |
| <b>626</b> | 8.8           | 0.600            | 0.29        | 2.20           | 0.098     | 5.0                 |
| <b>628</b> | 8.7           | 0.540            | 0.26        | 2.50           | 0.097     | 7.0                 |
| <b>629</b> | 7.6           | 0.685            | 0.23        | 2.30           | 0.111     | 20.0                |
| <b>631</b> | 10.4          | 0.280            | 0.54        | 2.70           | 0.105     | 5.0                 |
| <b>632</b> | 7.6           | 0.410            | 0.14        | 3.00           | 0.087     | 21.0                |
| <b>633</b> | 10.1          | 0.935            | 0.22        | 3.40           | 0.105     | 11.0                |
| <b>634</b> | 7.9           | 0.350            | 0.21        | 1.90           | 0.073     | 46.0                |
| <b>635</b> | 8.7           | 0.840            | 0.00        | 1.40           | 0.065     | 24.0                |
| <b>636</b> | 9.6           | 0.880            | 0.28        | 2.40           | 0.086     | 30.0                |
| <b>637</b> | 9.5           | 0.885            | 0.27        | 2.30           | 0.084     | 31.0                |
| <b>638</b> | 7.7           | 0.915            | 0.12        | 2.20           | 0.143     | 7.0                 |
| <b>639</b> | 8.9           | 0.290            | 0.35        | 1.90           | 0.067     | 25.0                |
| <b>640</b> | 9.9           | 0.540            | 0.45        | 2.30           | 0.071     | 16.0                |
| <b>641</b> | 9.5           | 0.590            | 0.44        | 2.30           | 0.071     | 21.0                |
| <b>645</b> | 7.8           | 0.640            | 0.10        | 6.00           | 0.115     | 5.0                 |
| <b>646</b> | 7.3           | 0.670            | 0.05        | 3.60           | 0.107     | 6.0                 |
| <b>647</b> | 8.3           | 0.845            | 0.01        | 2.20           | 0.070     | 5.0                 |
| <b>648</b> | 8.7           | 0.480            | 0.30        | 2.80           | 0.066     | 10.0                |
| <b>649</b> | 6.7           | 0.420            | 0.27        | 8.60           | 0.068     | 24.0                |
| <b>650</b> | 10.7          | 0.430            | 0.39        | 2.20           | 0.106     | 8.0                 |
| <b>651</b> | 9.8           | 0.880            | 0.25        | 2.50           | 0.104     | 35.0                |
| <b>652</b> | 15.9          | 0.360            | 0.65        | 7.50           | 0.096     | 22.0                |
| <b>653</b> | 9.4           | 0.330            | 0.59        | 2.80           | 0.079     | 9.0                 |
| <b>654</b> | 8.6           | 0.470            | 0.47        | 2.40           | 0.074     | 7.0                 |
| <b>655</b> | 9.7           | 0.550            | 0.17        | 2.90           | 0.087     | 20.0                |
| <b>657</b> | 12.0          | 0.500            | 0.59        | 1.40           | 0.073     | 23.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>658</b> | 7.2           | 0.520            | 0.07        | 1.40           | 0.074     | 5.0                 |
| <b>659</b> | 7.1           | 0.840            | 0.02        | 4.40           | 0.096     | 5.0                 |
| <b>661</b> | 7.5           | 0.420            | 0.31        | 1.60           | 0.080     | 15.0                |
| <b>662</b> | 7.2           | 0.570            | 0.06        | 1.60           | 0.076     | 9.0                 |
| <b>663</b> | 10.1          | 0.280            | 0.46        | 1.80           | 0.050     | 5.0                 |
| <b>664</b> | 12.1          | 0.400            | 0.52        | 2.00           | 0.092     | 15.0                |
| <b>665</b> | 9.4           | 0.590            | 0.14        | 2.00           | 0.084     | 25.0                |
| <b>666</b> | 8.3           | 0.490            | 0.36        | 1.80           | 0.222     | 6.0                 |
| <b>667</b> | 11.3          | 0.340            | 0.45        | 2.00           | 0.082     | 6.0                 |
| <b>668</b> | 10.0          | 0.730            | 0.43        | 2.30           | 0.059     | 15.0                |
| <b>670</b> | 6.9           | 0.400            | 0.24        | 2.50           | 0.083     | 30.0                |
| <b>671</b> | 8.2           | 0.730            | 0.21        | 1.70           | 0.074     | 5.0                 |
| <b>672</b> | 9.8           | 1.240            | 0.34        | 2.00           | 0.079     | 32.0                |
| <b>674</b> | 10.8          | 0.400            | 0.41        | 2.20           | 0.084     | 7.0                 |
| <b>675</b> | 9.3           | 0.410            | 0.39        | 2.20           | 0.064     | 12.0                |
| <b>677</b> | 8.6           | 0.800            | 0.11        | 2.30           | 0.084     | 12.0                |
| <b>678</b> | 8.3           | 0.780            | 0.10        | 2.60           | 0.081     | 45.0                |
| <b>679</b> | 10.8          | 0.260            | 0.45        | 3.30           | 0.060     | 20.0                |
| <b>680</b> | 13.3          | 0.430            | 0.58        | 1.90           | 0.070     | 15.0                |
| <b>681</b> | 8.0           | 0.450            | 0.23        | 2.20           | 0.094     | 16.0                |
| <b>682</b> | 8.5           | 0.460            | 0.31        | 2.25           | 0.078     | 32.0                |
| <b>683</b> | 8.1           | 0.780            | 0.23        | 2.60           | 0.059     | 5.0                 |
| <b>684</b> | 9.8           | 0.980            | 0.32        | 2.30           | 0.078     | 35.0                |
| <b>686</b> | 7.1           | 0.650            | 0.18        | 1.80           | 0.070     | 13.0                |
| <b>687</b> | 9.1           | 0.640            | 0.23        | 3.10           | 0.095     | 13.0                |
| <b>688</b> | 7.7           | 0.660            | 0.04        | 1.60           | 0.039     | 4.0                 |
| <b>689</b> | 8.1           | 0.380            | 0.48        | 1.80           | 0.157     | 5.0                 |
| <b>690</b> | 7.4           | 1.185            | 0.00        | 4.25           | 0.097     | 5.0                 |
| <b>691</b> | 9.2           | 0.920            | 0.24        | 2.60           | 0.087     | 12.0                |
| <b>692</b> | 8.6           | 0.490            | 0.51        | 2.00           | 0.422     | 16.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>693</b> | 9.0           | 0.480            | 0.32        | 2.80           | 0.084     | 21.0                |
| <b>694</b> | 9.0           | 0.470            | 0.31        | 2.70           | 0.084     | 24.0                |
| <b>695</b> | 5.1           | 0.470            | 0.02        | 1.30           | 0.034     | 18.0                |
| <b>696</b> | 7.0           | 0.650            | 0.02        | 2.10           | 0.066     | 8.0                 |
| <b>698</b> | 9.4           | 0.615            | 0.28        | 3.20           | 0.087     | 18.0                |
| <b>699</b> | 11.8          | 0.380            | 0.55        | 2.10           | 0.071     | 5.0                 |
| <b>700</b> | 10.6          | 1.020            | 0.43        | 2.90           | 0.076     | 26.0                |
| <b>702</b> | 7.0           | 0.640            | 0.02        | 2.10           | 0.067     | 9.0                 |
| <b>703</b> | 7.5           | 0.380            | 0.48        | 2.60           | 0.073     | 22.0                |
| <b>704</b> | 9.1           | 0.765            | 0.04        | 1.60           | 0.078     | 4.0                 |
| <b>705</b> | 8.4           | 1.035            | 0.15        | 6.00           | 0.073     | 11.0                |
| <b>706</b> | 7.0           | 0.780            | 0.08        | 2.00           | 0.093     | 10.0                |
| <b>707</b> | 7.4           | 0.490            | 0.19        | 3.00           | 0.077     | 16.0                |
| <b>708</b> | 7.8           | 0.545            | 0.12        | 2.50           | 0.068     | 11.0                |
| <b>709</b> | 9.7           | 0.310            | 0.47        | 1.60           | 0.062     | 13.0                |
| <b>710</b> | 10.6          | 1.025            | 0.43        | 2.80           | 0.080     | 21.0                |
| <b>711</b> | 8.9           | 0.565            | 0.34        | 3.00           | 0.093     | 16.0                |
| <b>712</b> | 8.7           | 0.690            | 0.00        | 3.20           | 0.084     | 13.0                |
| <b>713</b> | 8.0           | 0.430            | 0.36        | 2.30           | 0.075     | 10.0                |
| <b>714</b> | 9.9           | 0.740            | 0.28        | 2.60           | 0.078     | 21.0                |
| <b>715</b> | 7.2           | 0.490            | 0.18        | 2.70           | 0.069     | 13.0                |
| <b>717</b> | 7.6           | 0.460            | 0.11        | 2.60           | 0.079     | 12.0                |
| <b>718</b> | 8.4           | 0.560            | 0.04        | 2.00           | 0.082     | 10.0                |
| <b>719</b> | 7.1           | 0.660            | 0.00        | 3.90           | 0.086     | 17.0                |
| <b>721</b> | 8.9           | 0.480            | 0.24        | 2.85           | 0.094     | 35.0                |
| <b>722</b> | 7.6           | 0.420            | 0.08        | 2.70           | 0.084     | 15.0                |
| <b>723</b> | 7.1           | 0.310            | 0.30        | 2.20           | 0.053     | 36.0                |
| <b>724</b> | 7.5           | 1.115            | 0.10        | 3.10           | 0.086     | 5.0                 |
| <b>725</b> | 9.0           | 0.660            | 0.17        | 3.00           | 0.077     | 5.0                 |
| <b>726</b> | 8.1           | 0.720            | 0.09        | 2.80           | 0.084     | 18.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>727</b> | 6.4           | 0.570            | 0.02        | 1.80           | 0.067     | 4.0                 |
| <b>729</b> | 6.4           | 0.865            | 0.03        | 3.20           | 0.071     | 27.0                |
| <b>730</b> | 9.5           | 0.550            | 0.66        | 2.30           | 0.387     | 12.0                |
| <b>731</b> | 8.9           | 0.875            | 0.13        | 3.45           | 0.088     | 4.0                 |
| <b>732</b> | 7.3           | 0.835            | 0.03        | 2.10           | 0.092     | 10.0                |
| <b>733</b> | 7.0           | 0.450            | 0.34        | 2.70           | 0.082     | 16.0                |
| <b>734</b> | 7.7           | 0.560            | 0.20        | 2.00           | 0.075     | 9.0                 |
| <b>735</b> | 7.7           | 0.965            | 0.10        | 2.10           | 0.112     | 11.0                |
| <b>737</b> | 8.2           | 0.590            | 0.00        | 2.50           | 0.093     | 19.0                |
| <b>738</b> | 9.0           | 0.460            | 0.23        | 2.80           | 0.092     | 28.0                |
| <b>739</b> | 9.0           | 0.690            | 0.00        | 2.40           | 0.088     | 19.0                |
| <b>740</b> | 8.3           | 0.760            | 0.29        | 4.20           | 0.075     | 12.0                |
| <b>741</b> | 9.2           | 0.530            | 0.24        | 2.60           | 0.078     | 28.0                |
| <b>742</b> | 6.5           | 0.615            | 0.00        | 1.90           | 0.065     | 9.0                 |
| <b>743</b> | 11.6          | 0.410            | 0.58        | 2.80           | 0.096     | 25.0                |
| <b>744</b> | 11.1          | 0.390            | 0.54        | 2.70           | 0.095     | 21.0                |
| <b>745</b> | 7.3           | 0.510            | 0.18        | 2.10           | 0.070     | 12.0                |
| <b>746</b> | 8.2           | 0.340            | 0.38        | 2.50           | 0.080     | 12.0                |
| <b>747</b> | 8.6           | 0.330            | 0.40        | 2.60           | 0.083     | 16.0                |
| <b>748</b> | 7.2           | 0.500            | 0.18        | 2.10           | 0.071     | 12.0                |
| <b>750</b> | 8.3           | 0.650            | 0.10        | 2.90           | 0.089     | 17.0                |
| <b>752</b> | 7.6           | 0.540            | 0.13        | 2.50           | 0.097     | 24.0                |
| <b>754</b> | 7.8           | 0.480            | 0.68        | 1.70           | 0.415     | 14.0                |
| <b>755</b> | 7.8           | 0.910            | 0.07        | 1.90           | 0.058     | 22.0                |
| <b>756</b> | 6.3           | 0.980            | 0.01        | 2.00           | 0.057     | 15.0                |
| <b>757</b> | 8.1           | 0.870            | 0.00        | 2.20           | 0.084     | 10.0                |
| <b>759</b> | 8.8           | 0.420            | 0.21        | 2.50           | 0.092     | 33.0                |
| <b>760</b> | 9.0           | 0.580            | 0.25        | 2.80           | 0.075     | 9.0                 |
| <b>761</b> | 9.3           | 0.655            | 0.26        | 2.00           | 0.096     | 5.0                 |
| <b>762</b> | 8.8           | 0.700            | 0.00        | 1.70           | 0.069     | 8.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>764</b> | 9.1           | 0.680            | 0.11        | 2.80           | 0.093     | 11.0                |
| <b>765</b> | 9.2           | 0.670            | 0.10        | 3.00           | 0.091     | 12.0                |
| <b>766</b> | 8.8           | 0.590            | 0.18        | 2.90           | 0.089     | 12.0                |
| <b>767</b> | 7.5           | 0.600            | 0.32        | 2.70           | 0.103     | 13.0                |
| <b>768</b> | 7.1           | 0.590            | 0.02        | 2.30           | 0.082     | 24.0                |
| <b>769</b> | 7.9           | 0.720            | 0.01        | 1.90           | 0.076     | 7.0                 |
| <b>771</b> | 9.4           | 0.685            | 0.26        | 2.40           | 0.082     | 23.0                |
| <b>772</b> | 9.5           | 0.570            | 0.27        | 2.30           | 0.082     | 23.0                |
| <b>773</b> | 7.9           | 0.400            | 0.29        | 1.80           | 0.157     | 1.0                 |
| <b>774</b> | 7.9           | 0.400            | 0.30        | 1.80           | 0.157     | 2.0                 |
| <b>775</b> | 7.2           | 1.000            | 0.00        | 3.00           | 0.102     | 7.0                 |
| <b>776</b> | 6.9           | 0.765            | 0.18        | 2.40           | 0.243     | 5.5                 |
| <b>777</b> | 6.9           | 0.635            | 0.17        | 2.40           | 0.241     | 6.0                 |
| <b>778</b> | 8.3           | 0.430            | 0.30        | 3.40           | 0.079     | 7.0                 |
| <b>779</b> | 7.1           | 0.520            | 0.03        | 2.60           | 0.076     | 21.0                |
| <b>780</b> | 7.0           | 0.570            | 0.00        | 2.00           | 0.190     | 12.0                |
| <b>781</b> | 6.5           | 0.460            | 0.14        | 2.40           | 0.114     | 9.0                 |
| <b>782</b> | 9.0           | 0.820            | 0.05        | 2.40           | 0.081     | 26.0                |
| <b>784</b> | 7.1           | 0.590            | 0.01        | 2.50           | 0.077     | 20.0                |
| <b>785</b> | 9.9           | 0.350            | 0.41        | 2.30           | 0.083     | 11.0                |
| <b>787</b> | 10.0          | 0.560            | 0.24        | 2.20           | 0.079     | 19.0                |
| <b>789</b> | 8.6           | 0.630            | 0.17        | 2.90           | 0.099     | 21.0                |
| <b>790</b> | 7.4           | 0.370            | 0.43        | 2.60           | 0.082     | 18.0                |
| <b>791</b> | 8.8           | 0.640            | 0.17        | 2.90           | 0.084     | 25.0                |
| <b>792</b> | 7.1           | 0.610            | 0.02        | 2.50           | 0.081     | 17.0                |
| <b>793</b> | 7.7           | 0.600            | 0.00        | 2.60           | 0.055     | 7.0                 |
| <b>794</b> | 10.1          | 0.270            | 0.54        | 2.30           | 0.065     | 7.0                 |
| <b>795</b> | 10.8          | 0.890            | 0.30        | 2.60           | 0.132     | 7.0                 |
| <b>796</b> | 8.7           | 0.460            | 0.31        | 2.50           | 0.126     | 24.0                |
| <b>797</b> | 9.3           | 0.370            | 0.44        | 1.60           | 0.038     | 21.0                |



|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>798</b> | 9.4           | 0.500            | 0.34        | 3.60           | 0.082     | 5.0                 |
| <b>800</b> | 7.2           | 0.610            | 0.08        | 4.00           | 0.082     | 26.0                |
| <b>801</b> | 8.6           | 0.550            | 0.09        | 3.30           | 0.068     | 8.0                 |
| <b>802</b> | 5.1           | 0.585            | 0.00        | 1.70           | 0.044     | 14.0                |
| <b>803</b> | 7.7           | 0.560            | 0.08        | 2.50           | 0.114     | 14.0                |
| <b>804</b> | 8.4           | 0.520            | 0.22        | 2.70           | 0.084     | 4.0                 |
| <b>805</b> | 8.2           | 0.280            | 0.40        | 2.40           | 0.052     | 4.0                 |
| <b>806</b> | 8.4           | 0.250            | 0.39        | 2.00           | 0.041     | 4.0                 |
| <b>808</b> | 7.4           | 0.530            | 0.12        | 1.90           | 0.165     | 4.0                 |
| <b>809</b> | 7.6           | 0.480            | 0.31        | 2.80           | 0.070     | 4.0                 |
| <b>810</b> | 7.3           | 0.490            | 0.10        | 2.60           | 0.068     | 4.0                 |
| <b>811</b> | 12.9          | 0.500            | 0.55        | 2.80           | 0.072     | 7.0                 |
| <b>812</b> | 10.8          | 0.450            | 0.33        | 2.50           | 0.099     | 20.0                |
| <b>813</b> | 6.9           | 0.390            | 0.24        | 2.10           | 0.102     | 4.0                 |
| <b>814</b> | 12.6          | 0.410            | 0.54        | 2.80           | 0.103     | 19.0                |
| <b>816</b> | 9.8           | 0.510            | 0.19        | 3.20           | 0.081     | 8.0                 |
| <b>817</b> | 10.8          | 0.290            | 0.42        | 1.60           | 0.084     | 19.0                |
| <b>818</b> | 7.1           | 0.715            | 0.00        | 2.35           | 0.071     | 21.0                |
| <b>819</b> | 9.1           | 0.660            | 0.15        | 3.20           | 0.097     | 9.0                 |
| <b>820</b> | 7.0           | 0.685            | 0.00        | 1.90           | 0.099     | 9.0                 |
| <b>821</b> | 4.9           | 0.420            | 0.00        | 2.10           | 0.048     | 16.0                |
| <b>822</b> | 6.7           | 0.540            | 0.13        | 2.00           | 0.076     | 15.0                |
| <b>824</b> | 7.1           | 0.480            | 0.28        | 2.80           | 0.068     | 6.0                 |
| <b>825</b> | 7.1           | 0.460            | 0.14        | 2.80           | 0.076     | 15.0                |
| <b>826</b> | 7.5           | 0.270            | 0.34        | 2.30           | 0.050     | 4.0                 |
| <b>828</b> | 7.8           | 0.570            | 0.09        | 2.30           | 0.065     | 34.0                |
| <b>829</b> | 5.9           | 0.610            | 0.08        | 2.10           | 0.071     | 16.0                |
| <b>830</b> | 7.5           | 0.685            | 0.07        | 2.50           | 0.058     | 5.0                 |
| <b>832</b> | 10.4          | 0.440            | 0.42        | 1.50           | 0.145     | 34.0                |
| <b>833</b> | 11.6          | 0.470            | 0.44        | 1.60           | 0.147     | 36.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>834</b> | 8.8           | 0.685            | 0.26        | 1.60           | 0.088     | 16.0                |
| <b>835</b> | 7.6           | 0.665            | 0.10        | 1.50           | 0.066     | 27.0                |
| <b>836</b> | 6.7           | 0.280            | 0.28        | 2.40           | 0.012     | 36.0                |
| <b>838</b> | 10.1          | 0.310            | 0.35        | 1.60           | 0.075     | 9.0                 |
| <b>839</b> | 6.0           | 0.500            | 0.04        | 2.20           | 0.092     | 13.0                |
| <b>840</b> | 11.1          | 0.420            | 0.47        | 2.65           | 0.085     | 9.0                 |
| <b>841</b> | 6.6           | 0.660            | 0.00        | 3.00           | 0.115     | 21.0                |
| <b>842</b> | 10.6          | 0.500            | 0.45        | 2.60           | 0.119     | 34.0                |
| <b>843</b> | 7.1           | 0.685            | 0.35        | 2.00           | 0.088     | 9.0                 |
| <b>844</b> | 9.9           | 0.250            | 0.46        | 1.70           | 0.062     | 26.0                |
| <b>845</b> | 6.4           | 0.640            | 0.21        | 1.80           | 0.081     | 14.0                |
| <b>847</b> | 7.4           | 0.680            | 0.16        | 1.80           | 0.078     | 12.0                |
| <b>849</b> | 6.4           | 0.630            | 0.21        | 1.60           | 0.080     | 12.0                |
| <b>850</b> | 9.3           | 0.430            | 0.44        | 1.90           | 0.085     | 9.0                 |
| <b>852</b> | 8.0           | 0.420            | 0.32        | 2.50           | 0.080     | 26.0                |
| <b>853</b> | 9.3           | 0.360            | 0.39        | 1.50           | 0.080     | 41.0                |
| <b>855</b> | 7.6           | 0.735            | 0.02        | 2.50           | 0.071     | 10.0                |
| <b>857</b> | 8.2           | 0.260            | 0.34        | 2.50           | 0.073     | 16.0                |
| <b>858</b> | 11.7          | 0.280            | 0.47        | 1.70           | 0.054     | 17.0                |
| <b>859</b> | 6.8           | 0.560            | 0.22        | 1.80           | 0.074     | 15.0                |
| <b>860</b> | 7.2           | 0.620            | 0.06        | 2.70           | 0.077     | 15.0                |
| <b>861</b> | 5.8           | 1.010            | 0.66        | 2.00           | 0.039     | 15.0                |
| <b>862</b> | 7.5           | 0.420            | 0.32        | 2.70           | 0.067     | 7.0                 |
| <b>863</b> | 7.2           | 0.620            | 0.06        | 2.50           | 0.078     | 17.0                |
| <b>865</b> | 7.2           | 0.635            | 0.07        | 2.60           | 0.077     | 16.0                |
| <b>866</b> | 6.8           | 0.490            | 0.22        | 2.30           | 0.071     | 13.0                |
| <b>867</b> | 6.9           | 0.510            | 0.23        | 2.00           | 0.072     | 13.0                |
| <b>869</b> | 7.6           | 0.630            | 0.03        | 2.00           | 0.080     | 27.0                |
| <b>870</b> | 7.7           | 0.715            | 0.01        | 2.10           | 0.064     | 31.0                |
| <b>871</b> | 6.9           | 0.560            | 0.03        | 1.50           | 0.086     | 36.0                |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>872</b> | 7.3           | 0.350            | 0.24        | 2.00           | 0.067     | 28.0                |
| <b>873</b> | 9.1           | 0.210            | 0.37        | 1.60           | 0.067     | 6.0                 |
| <b>874</b> | 10.4          | 0.380            | 0.46        | 2.10           | 0.104     | 6.0                 |
| <b>875</b> | 8.8           | 0.310            | 0.40        | 2.80           | 0.109     | 7.0                 |
| <b>876</b> | 7.1           | 0.470            | 0.00        | 2.20           | 0.067     | 7.0                 |
| <b>878</b> | 8.8           | 0.610            | 0.19        | 4.00           | 0.094     | 30.0                |
| <b>879</b> | 7.2           | 0.600            | 0.04        | 2.50           | 0.076     | 18.0                |
| <b>880</b> | 9.2           | 0.560            | 0.18        | 1.60           | 0.078     | 10.0                |
| <b>881</b> | 7.6           | 0.715            | 0.00        | 2.10           | 0.068     | 30.0                |
| <b>882</b> | 8.4           | 0.310            | 0.29        | 3.10           | 0.194     | 14.0                |
| <b>885</b> | 8.9           | 0.750            | 0.14        | 2.50           | 0.086     | 9.0                 |
| <b>886</b> | 9.0           | 0.800            | 0.12        | 2.40           | 0.083     | 8.0                 |
| <b>887</b> | 10.7          | 0.520            | 0.38        | 2.60           | 0.066     | 29.0                |
| <b>888</b> | 6.8           | 0.570            | 0.00        | 2.50           | 0.072     | 32.0                |
| <b>889</b> | 10.7          | 0.900            | 0.34        | 6.60           | 0.112     | 23.0                |
| <b>890</b> | 7.2           | 0.340            | 0.24        | 2.00           | 0.071     | 30.0                |
| <b>891</b> | 7.2           | 0.660            | 0.03        | 2.30           | 0.078     | 16.0                |
| <b>892</b> | 10.1          | 0.450            | 0.23        | 1.90           | 0.082     | 10.0                |
| <b>894</b> | 7.2           | 0.630            | 0.03        | 2.20           | 0.080     | 17.0                |
| <b>895</b> | 7.1           | 0.590            | 0.01        | 2.30           | 0.080     | 27.0                |
| <b>896</b> | 8.3           | 0.310            | 0.39        | 2.40           | 0.078     | 17.0                |
| <b>899</b> | 8.3           | 1.020            | 0.02        | 3.40           | 0.084     | 6.0                 |
| <b>900</b> | 8.9           | 0.310            | 0.36        | 2.60           | 0.056     | 10.0                |
| <b>901</b> | 7.4           | 0.635            | 0.10        | 2.40           | 0.080     | 16.0                |
| <b>903</b> | 6.8           | 0.590            | 0.06        | 6.00           | 0.060     | 11.0                |
| <b>905</b> | 9.2           | 0.580            | 0.20        | 3.00           | 0.081     | 15.0                |
| <b>906</b> | 7.2           | 0.540            | 0.27        | 2.60           | 0.084     | 12.0                |
| <b>907</b> | 6.1           | 0.560            | 0.00        | 2.20           | 0.079     | 6.0                 |
| <b>908</b> | 7.4           | 0.520            | 0.13        | 2.40           | 0.078     | 34.0                |
| <b>909</b> | 7.3           | 0.305            | 0.39        | 1.20           | 0.059     | 7.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>910</b> | 9.3           | 0.380            | 0.48        | 3.80           | 0.132     | 3.0                 |
| <b>911</b> | 9.1           | 0.280            | 0.46        | 9.00           | 0.114     | 3.0                 |
| <b>912</b> | 10.0          | 0.460            | 0.44        | 2.90           | 0.065     | 4.0                 |
| <b>913</b> | 9.4           | 0.395            | 0.46        | 4.60           | 0.094     | 3.0                 |
| <b>915</b> | 8.6           | 0.315            | 0.40        | 2.20           | 0.079     | 3.0                 |
| <b>916</b> | 5.3           | 0.715            | 0.19        | 1.50           | 0.161     | 7.0                 |
| <b>917</b> | 6.8           | 0.410            | 0.31        | 8.80           | 0.084     | 26.0                |
| <b>918</b> | 8.4           | 0.360            | 0.32        | 2.20           | 0.081     | 32.0                |
| <b>919</b> | 8.4           | 0.620            | 0.12        | 1.80           | 0.072     | 38.0                |
| <b>920</b> | 9.6           | 0.410            | 0.37        | 2.30           | 0.091     | 10.0                |
| <b>924</b> | 8.6           | 0.470            | 0.27        | 2.30           | 0.055     | 14.0                |
| <b>925</b> | 8.6           | 0.220            | 0.36        | 1.90           | 0.064     | 53.0                |
| <b>926</b> | 9.4           | 0.240            | 0.33        | 2.30           | 0.061     | 52.0                |
| <b>927</b> | 8.4           | 0.670            | 0.19        | 2.20           | 0.093     | 11.0                |
| <b>929</b> | 8.7           | 0.330            | 0.38        | 3.30           | 0.063     | 10.0                |
| <b>930</b> | 6.6           | 0.610            | 0.01        | 1.90           | 0.080     | 8.0                 |
| <b>931</b> | 7.4           | 0.610            | 0.01        | 2.00           | 0.074     | 13.0                |
| <b>932</b> | 7.6           | 0.400            | 0.29        | 1.90           | 0.078     | 29.0                |
| <b>935</b> | 8.8           | 0.300            | 0.38        | 2.30           | 0.060     | 19.0                |
| <b>937</b> | 12.0          | 0.630            | 0.50        | 1.40           | 0.071     | 6.0                 |
| <b>938</b> | 7.2           | 0.380            | 0.38        | 2.80           | 0.068     | 23.0                |
| <b>939</b> | 6.2           | 0.460            | 0.17        | 1.60           | 0.073     | 7.0                 |
| <b>940</b> | 9.6           | 0.330            | 0.52        | 2.20           | 0.074     | 13.0                |
| <b>941</b> | 9.9           | 0.270            | 0.49        | 5.00           | 0.082     | 9.0                 |
| <b>942</b> | 10.1          | 0.430            | 0.40        | 2.60           | 0.092     | 13.0                |
| <b>943</b> | 9.8           | 0.500            | 0.34        | 2.30           | 0.094     | 10.0                |
| <b>944</b> | 8.3           | 0.300            | 0.49        | 3.80           | 0.090     | 11.0                |
| <b>945</b> | 10.2          | 0.440            | 0.42        | 2.00           | 0.071     | 7.0                 |
| <b>946</b> | 10.2          | 0.440            | 0.58        | 4.10           | 0.092     | 11.0                |
| <b>947</b> | 8.3           | 0.280            | 0.48        | 2.10           | 0.093     | 6.0                 |

|            | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>948</b> | 8.9           | 0.120            | 0.45        | 1.80           | 0.075     | 10.0                |
| <b>952</b> | 8.2           | 0.310            | 0.40        | 2.20           | 0.058     | 6.0                 |
| <b>953</b> | 10.2          | 0.340            | 0.48        | 2.10           | 0.052     | 5.0                 |
| <b>954</b> | 7.6           | 0.430            | 0.40        | 2.70           | 0.082     | 6.0                 |
| <b>955</b> | 8.5           | 0.210            | 0.52        | 1.90           | 0.090     | 9.0                 |
| <b>956</b> | 9.0           | 0.360            | 0.52        | 2.10           | 0.111     | 5.0                 |
| <b>957</b> | 9.5           | 0.370            | 0.52        | 2.00           | 0.088     | 12.0                |
| <b>958</b> | 6.4           | 0.570            | 0.12        | 2.30           | 0.120     | 25.0                |
| <b>959</b> | 8.0           | 0.590            | 0.05        | 2.00           | 0.089     | 12.0                |
| <b>960</b> | 8.5           | 0.470            | 0.27        | 1.90           | 0.058     | 18.0                |
| <b>961</b> | 7.1           | 0.560            | 0.14        | 1.60           | 0.078     | 7.0                 |
| <b>962</b> | 6.6           | 0.570            | 0.02        | 2.10           | 0.115     | 6.0                 |
| <b>963</b> | 8.8           | 0.270            | 0.39        | 2.00           | 0.100     | 20.0                |
| <b>965</b> | 8.3           | 0.340            | 0.40        | 2.40           | 0.065     | 24.0                |
| <b>966</b> | 9.0           | 0.380            | 0.41        | 2.40           | 0.103     | 6.0                 |
| <b>967</b> | 8.5           | 0.660            | 0.20        | 2.10           | 0.097     | 23.0                |
| <b>968</b> | 9.0           | 0.400            | 0.43        | 2.40           | 0.068     | 29.0                |
| <b>969</b> | 6.7           | 0.560            | 0.09        | 2.90           | 0.079     | 7.0                 |
| <b>970</b> | 10.4          | 0.260            | 0.48        | 1.90           | 0.066     | 6.0                 |
| <b>972</b> | 10.1          | 0.380            | 0.50        | 2.40           | 0.104     | 6.0                 |
| <b>973</b> | 8.5           | 0.340            | 0.44        | 1.70           | 0.079     | 6.0                 |
| <b>974</b> | 8.8           | 0.330            | 0.41        | 5.90           | 0.073     | 7.0                 |
| <b>975</b> | 7.2           | 0.410            | 0.30        | 2.10           | 0.083     | 35.0                |
| <b>977</b> | 8.4           | 0.590            | 0.29        | 2.60           | 0.109     | 31.0                |
| <b>978</b> | 7.0           | 0.400            | 0.32        | 3.60           | 0.061     | 9.0                 |
| <b>979</b> | 12.2          | 0.450            | 0.49        | 1.40           | 0.075     | 3.0                 |
| <b>980</b> | 9.1           | 0.500            | 0.30        | 1.90           | 0.065     | 8.0                 |
| <b>981</b> | 9.5           | 0.860            | 0.26        | 1.90           | 0.079     | 13.0                |
| <b>982</b> | 7.3           | 0.520            | 0.32        | 2.10           | 0.070     | 51.0                |
| <b>985</b> | 7.4           | 0.580            | 0.00        | 2.00           | 0.064     | 7.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>986</b>  | 9.8           | 0.340            | 0.39        | 1.40           | 0.066     | 3.0                 |
| <b>987</b>  | 7.1           | 0.360            | 0.30        | 1.60           | 0.080     | 35.0                |
| <b>988</b>  | 7.7           | 0.390            | 0.12        | 1.70           | 0.097     | 19.0                |
| <b>989</b>  | 9.7           | 0.295            | 0.40        | 1.50           | 0.073     | 14.0                |
| <b>991</b>  | 7.1           | 0.340            | 0.28        | 2.00           | 0.082     | 31.0                |
| <b>992</b>  | 6.5           | 0.400            | 0.10        | 2.00           | 0.076     | 30.0                |
| <b>994</b>  | 10.0          | 0.350            | 0.45        | 2.50           | 0.092     | 20.0                |
| <b>995</b>  | 7.7           | 0.600            | 0.06        | 2.00           | 0.079     | 19.0                |
| <b>996</b>  | 5.6           | 0.660            | 0.00        | 2.20           | 0.087     | 3.0                 |
| <b>998</b>  | 8.9           | 0.840            | 0.34        | 1.40           | 0.050     | 4.0                 |
| <b>999</b>  | 6.4           | 0.690            | 0.00        | 1.65           | 0.055     | 7.0                 |
| <b>1000</b> | 7.5           | 0.430            | 0.30        | 2.20           | 0.062     | 6.0                 |
| <b>1001</b> | 9.9           | 0.350            | 0.38        | 1.50           | 0.058     | 31.0                |
| <b>1002</b> | 9.1           | 0.290            | 0.33        | 2.05           | 0.063     | 13.0                |
| <b>1003</b> | 6.8           | 0.360            | 0.32        | 1.80           | 0.067     | 4.0                 |
| <b>1004</b> | 8.2           | 0.430            | 0.29        | 1.60           | 0.081     | 27.0                |
| <b>1007</b> | 9.1           | 0.300            | 0.34        | 2.00           | 0.064     | 12.0                |
| <b>1008</b> | 8.9           | 0.350            | 0.40        | 3.60           | 0.110     | 12.0                |
| <b>1009</b> | 9.6           | 0.500            | 0.36        | 2.80           | 0.116     | 26.0                |
| <b>1010</b> | 8.9           | 0.280            | 0.45        | 1.70           | 0.067     | 7.0                 |
| <b>1011</b> | 8.9           | 0.320            | 0.31        | 2.00           | 0.088     | 12.0                |
| <b>1012</b> | 7.7           | 1.005            | 0.15        | 2.10           | 0.102     | 11.0                |
| <b>1013</b> | 7.5           | 0.710            | 0.00        | 1.60           | 0.092     | 22.0                |
| <b>1014</b> | 8.0           | 0.580            | 0.16        | 2.00           | 0.120     | 3.0                 |
| <b>1015</b> | 10.5          | 0.390            | 0.46        | 2.20           | 0.075     | 14.0                |
| <b>1016</b> | 8.9           | 0.380            | 0.40        | 2.20           | 0.068     | 12.0                |
| <b>1017</b> | 8.0           | 0.180            | 0.37        | 0.90           | 0.049     | 36.0                |
| <b>1019</b> | 7.0           | 0.500            | 0.14        | 1.80           | 0.078     | 10.0                |
| <b>1020</b> | 11.3          | 0.360            | 0.66        | 2.40           | 0.123     | 3.0                 |
| <b>1022</b> | 7.0           | 0.510            | 0.09        | 2.10           | 0.062     | 4.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1023</b> | 8.2           | 0.320            | 0.42        | 2.30           | 0.098     | 3.0                 |
| <b>1024</b> | 7.7           | 0.580            | 0.01        | 1.80           | 0.088     | 12.0                |
| <b>1025</b> | 8.6           | 0.830            | 0.00        | 2.80           | 0.095     | 17.0                |
| <b>1026</b> | 7.9           | 0.310            | 0.32        | 1.90           | 0.066     | 14.0                |
| <b>1027</b> | 6.4           | 0.795            | 0.00        | 2.20           | 0.065     | 28.0                |
| <b>1028</b> | 7.2           | 0.340            | 0.21        | 2.50           | 0.075     | 41.0                |
| <b>1030</b> | 7.1           | 0.590            | 0.00        | 2.10           | 0.091     | 9.0                 |
| <b>1031</b> | 7.3           | 0.550            | 0.01        | 1.80           | 0.093     | 9.0                 |
| <b>1032</b> | 8.1           | 0.820            | 0.00        | 4.10           | 0.095     | 5.0                 |
| <b>1033</b> | 7.5           | 0.570            | 0.08        | 2.60           | 0.089     | 14.0                |
| <b>1034</b> | 8.9           | 0.745            | 0.18        | 2.50           | 0.077     | 15.0                |
| <b>1035</b> | 10.1          | 0.370            | 0.34        | 2.40           | 0.085     | 5.0                 |
| <b>1036</b> | 7.6           | 0.310            | 0.34        | 2.50           | 0.082     | 26.0                |
| <b>1037</b> | 7.3           | 0.910            | 0.10        | 1.80           | 0.074     | 20.0                |
| <b>1038</b> | 8.7           | 0.410            | 0.41        | 6.20           | 0.078     | 25.0                |
| <b>1039</b> | 8.9           | 0.500            | 0.21        | 2.20           | 0.088     | 21.0                |
| <b>1040</b> | 7.4           | 0.965            | 0.00        | 2.20           | 0.088     | 16.0                |
| <b>1041</b> | 6.9           | 0.490            | 0.19        | 1.70           | 0.079     | 13.0                |
| <b>1043</b> | 9.5           | 0.390            | 0.41        | 8.90           | 0.069     | 18.0                |
| <b>1044</b> | 6.4           | 0.390            | 0.33        | 3.30           | 0.046     | 12.0                |
| <b>1045</b> | 6.9           | 0.440            | 0.00        | 1.40           | 0.070     | 32.0                |
| <b>1046</b> | 7.6           | 0.780            | 0.00        | 1.70           | 0.076     | 33.0                |
| <b>1047</b> | 7.1           | 0.430            | 0.17        | 1.80           | 0.082     | 27.0                |
| <b>1048</b> | 9.3           | 0.490            | 0.36        | 1.70           | 0.081     | 3.0                 |
| <b>1049</b> | 9.3           | 0.500            | 0.36        | 1.80           | 0.084     | 6.0                 |
| <b>1051</b> | 8.5           | 0.460            | 0.59        | 1.40           | 0.414     | 16.0                |
| <b>1052</b> | 5.6           | 0.605            | 0.05        | 2.40           | 0.073     | 19.0                |
| <b>1053</b> | 8.3           | 0.330            | 0.42        | 2.30           | 0.070     | 9.0                 |
| <b>1054</b> | 8.2           | 0.640            | 0.27        | 2.00           | 0.095     | 5.0                 |
| <b>1056</b> | 8.9           | 0.480            | 0.53        | 4.00           | 0.101     | 3.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1057</b> | 7.6           | 0.420            | 0.25        | 3.90           | 0.104     | 28.0                |
| <b>1058</b> | 9.9           | 0.530            | 0.57        | 2.40           | 0.093     | 30.0                |
| <b>1060</b> | 11.6          | 0.230            | 0.57        | 1.80           | 0.074     | 3.0                 |
| <b>1061</b> | 9.1           | 0.400            | 0.50        | 1.80           | 0.071     | 7.0                 |
| <b>1062</b> | 8.0           | 0.380            | 0.44        | 1.90           | 0.098     | 6.0                 |
| <b>1063</b> | 10.2          | 0.290            | 0.65        | 2.40           | 0.075     | 6.0                 |
| <b>1064</b> | 8.2           | 0.740            | 0.09        | 2.00           | 0.067     | 5.0                 |
| <b>1065</b> | 7.7           | 0.610            | 0.18        | 2.40           | 0.083     | 6.0                 |
| <b>1066</b> | 6.6           | 0.520            | 0.08        | 2.40           | 0.070     | 13.0                |
| <b>1067</b> | 11.1          | 0.310            | 0.53        | 2.20           | 0.060     | 3.0                 |
| <b>1069</b> | 8.0           | 0.620            | 0.35        | 2.80           | 0.086     | 28.0                |
| <b>1070</b> | 9.3           | 0.330            | 0.45        | 1.50           | 0.057     | 19.0                |
| <b>1071</b> | 7.5           | 0.770            | 0.20        | 8.10           | 0.098     | 30.0                |
| <b>1072</b> | 7.2           | 0.350            | 0.26        | 1.80           | 0.083     | 33.0                |
| <b>1073</b> | 8.0           | 0.620            | 0.33        | 2.70           | 0.088     | 16.0                |
| <b>1075</b> | 9.1           | 0.250            | 0.34        | 2.00           | 0.071     | 45.0                |
| <b>1076</b> | 9.9           | 0.320            | 0.56        | 2.00           | 0.073     | 3.0                 |
| <b>1077</b> | 8.6           | 0.370            | 0.65        | 6.40           | 0.080     | 3.0                 |
| <b>1079</b> | 7.9           | 0.300            | 0.68        | 8.30           | 0.050     | 37.5                |
| <b>1080</b> | 10.3          | 0.270            | 0.56        | 1.40           | 0.047     | 3.0                 |
| <b>1081</b> | 7.9           | 0.300            | 0.68        | 8.30           | 0.050     | 37.5                |
| <b>1082</b> | 7.2           | 0.380            | 0.30        | 1.80           | 0.073     | 31.0                |
| <b>1083</b> | 8.7           | 0.420            | 0.45        | 2.40           | 0.072     | 32.0                |
| <b>1085</b> | 6.8           | 0.480            | 0.08        | 1.80           | 0.074     | 40.0                |
| <b>1086</b> | 8.5           | 0.340            | 0.40        | 4.70           | 0.055     | 3.0                 |
| <b>1087</b> | 7.9           | 0.190            | 0.42        | 1.60           | 0.057     | 18.0                |
| <b>1088</b> | 11.6          | 0.410            | 0.54        | 1.50           | 0.095     | 22.0                |
| <b>1090</b> | 10.0          | 0.260            | 0.54        | 1.90           | 0.083     | 42.0                |
| <b>1091</b> | 7.9           | 0.340            | 0.42        | 2.00           | 0.086     | 8.0                 |
| <b>1092</b> | 7.0           | 0.540            | 0.09        | 2.00           | 0.081     | 10.0                |



|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1093</b> | 9.2           | 0.310            | 0.36        | 2.20           | 0.079     | 11.0                |
| <b>1094</b> | 6.6           | 0.725            | 0.09        | 5.50           | 0.117     | 9.0                 |
| <b>1095</b> | 9.4           | 0.400            | 0.47        | 2.50           | 0.087     | 6.0                 |
| <b>1097</b> | 8.6           | 0.520            | 0.38        | 1.50           | 0.096     | 5.0                 |
| <b>1098</b> | 8.0           | 0.310            | 0.45        | 2.10           | 0.216     | 5.0                 |
| <b>1100</b> | 8.4           | 0.340            | 0.42        | 2.10           | 0.072     | 23.0                |
| <b>1101</b> | 7.4           | 0.490            | 0.27        | 2.10           | 0.071     | 14.0                |
| <b>1102</b> | 6.1           | 0.480            | 0.09        | 1.70           | 0.078     | 18.0                |
| <b>1104</b> | 8.0           | 0.480            | 0.34        | 2.20           | 0.073     | 16.0                |
| <b>1105</b> | 6.3           | 0.570            | 0.28        | 2.10           | 0.048     | 13.0                |
| <b>1106</b> | 8.2           | 0.230            | 0.42        | 1.90           | 0.069     | 9.0                 |
| <b>1107</b> | 9.1           | 0.300            | 0.41        | 2.00           | 0.068     | 10.0                |
| <b>1108</b> | 8.1           | 0.780            | 0.10        | 3.30           | 0.090     | 4.0                 |
| <b>1109</b> | 10.8          | 0.470            | 0.43        | 2.10           | 0.171     | 27.0                |
| <b>1110</b> | 8.3           | 0.530            | 0.00        | 1.40           | 0.070     | 6.0                 |
| <b>1111</b> | 5.4           | 0.420            | 0.27        | 2.00           | 0.092     | 23.0                |
| <b>1112</b> | 7.9           | 0.330            | 0.41        | 1.50           | 0.056     | 6.0                 |
| <b>1113</b> | 8.9           | 0.240            | 0.39        | 1.60           | 0.074     | 3.0                 |
| <b>1114</b> | 5.0           | 0.400            | 0.50        | 4.30           | 0.046     | 29.0                |
| <b>1115</b> | 7.0           | 0.690            | 0.07        | 2.50           | 0.091     | 15.0                |
| <b>1118</b> | 7.1           | 0.390            | 0.12        | 2.10           | 0.065     | 14.0                |
| <b>1119</b> | 5.6           | 0.660            | 0.00        | 2.50           | 0.066     | 7.0                 |
| <b>1120</b> | 7.9           | 0.540            | 0.34        | 2.50           | 0.076     | 8.0                 |
| <b>1121</b> | 6.6           | 0.500            | 0.00        | 1.80           | 0.062     | 21.0                |
| <b>1122</b> | 6.3           | 0.470            | 0.00        | 1.40           | 0.055     | 27.0                |
| <b>1123</b> | 10.7          | 0.400            | 0.37        | 1.90           | 0.081     | 17.0                |
| <b>1124</b> | 6.5           | 0.580            | 0.00        | 2.20           | 0.096     | 3.0                 |
| <b>1125</b> | 8.8           | 0.240            | 0.35        | 1.70           | 0.055     | 13.0                |
| <b>1126</b> | 5.8           | 0.290            | 0.26        | 1.70           | 0.063     | 3.0                 |
| <b>1127</b> | 6.3           | 0.760            | 0.00        | 2.90           | 0.072     | 26.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1128</b> | 10.0          | 0.430            | 0.33        | 2.70           | 0.095     | 28.0                |
| <b>1129</b> | 10.5          | 0.430            | 0.35        | 3.30           | 0.092     | 24.0                |
| <b>1130</b> | 9.1           | 0.600            | 0.00        | 1.90           | 0.058     | 5.0                 |
| <b>1131</b> | 5.9           | 0.190            | 0.21        | 1.70           | 0.045     | 57.0                |
| <b>1132</b> | 7.4           | 0.360            | 0.34        | 1.80           | 0.075     | 18.0                |
| <b>1133</b> | 7.2           | 0.480            | 0.07        | 5.50           | 0.089     | 10.0                |
| <b>1134</b> | 8.5           | 0.280            | 0.35        | 1.70           | 0.061     | 6.0                 |
| <b>1135</b> | 8.0           | 0.250            | 0.43        | 1.70           | 0.067     | 22.0                |
| <b>1136</b> | 10.4          | 0.520            | 0.45        | 2.00           | 0.080     | 6.0                 |
| <b>1138</b> | 7.5           | 0.410            | 0.15        | 3.70           | 0.104     | 29.0                |
| <b>1139</b> | 8.2           | 0.510            | 0.24        | 2.00           | 0.079     | 16.0                |
| <b>1140</b> | 7.3           | 0.400            | 0.30        | 1.70           | 0.080     | 33.0                |
| <b>1141</b> | 8.2           | 0.380            | 0.32        | 2.50           | 0.080     | 24.0                |
| <b>1142</b> | 6.9           | 0.450            | 0.11        | 2.40           | 0.043     | 6.0                 |
| <b>1143</b> | 7.0           | 0.220            | 0.30        | 1.80           | 0.065     | 16.0                |
| <b>1144</b> | 7.3           | 0.320            | 0.23        | 2.30           | 0.066     | 35.0                |
| <b>1145</b> | 8.2           | 0.200            | 0.43        | 2.50           | 0.076     | 31.0                |
| <b>1146</b> | 7.8           | 0.500            | 0.12        | 1.80           | 0.178     | 6.0                 |
| <b>1147</b> | 10.0          | 0.410            | 0.45        | 6.20           | 0.071     | 6.0                 |
| <b>1148</b> | 7.8           | 0.390            | 0.42        | 2.00           | 0.086     | 9.0                 |
| <b>1149</b> | 10.0          | 0.350            | 0.47        | 2.00           | 0.061     | 6.0                 |
| <b>1150</b> | 8.2           | 0.330            | 0.32        | 2.80           | 0.067     | 4.0                 |
| <b>1151</b> | 6.1           | 0.580            | 0.23        | 2.50           | 0.044     | 16.0                |
| <b>1152</b> | 8.3           | 0.600            | 0.25        | 2.20           | 0.118     | 9.0                 |
| <b>1153</b> | 9.6           | 0.420            | 0.35        | 2.10           | 0.083     | 17.0                |
| <b>1154</b> | 6.6           | 0.580            | 0.00        | 2.20           | 0.100     | 50.0                |
| <b>1156</b> | 8.5           | 0.180            | 0.51        | 1.75           | 0.071     | 45.0                |
| <b>1157</b> | 5.1           | 0.510            | 0.18        | 2.10           | 0.042     | 16.0                |
| <b>1158</b> | 6.7           | 0.410            | 0.43        | 2.80           | 0.076     | 22.0                |
| <b>1159</b> | 10.2          | 0.410            | 0.43        | 2.20           | 0.110     | 11.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1160</b> | 10.6          | 0.360            | 0.57        | 2.30           | 0.087     | 6.0                 |
| <b>1161</b> | 8.8           | 0.450            | 0.43        | 1.40           | 0.076     | 12.0                |
| <b>1162</b> | 8.5           | 0.320            | 0.42        | 2.30           | 0.075     | 12.0                |
| <b>1163</b> | 9.0           | 0.785            | 0.24        | 1.70           | 0.078     | 10.0                |
| <b>1165</b> | 8.5           | 0.440            | 0.50        | 1.90           | 0.369     | 15.0                |
| <b>1166</b> | 9.9           | 0.540            | 0.26        | 2.00           | 0.111     | 7.0                 |
| <b>1167</b> | 8.2           | 0.330            | 0.39        | 2.50           | 0.074     | 29.0                |
| <b>1168</b> | 6.5           | 0.340            | 0.27        | 2.80           | 0.067     | 8.0                 |
| <b>1169</b> | 7.6           | 0.500            | 0.29        | 2.30           | 0.086     | 5.0                 |
| <b>1170</b> | 9.2           | 0.360            | 0.34        | 1.60           | 0.062     | 5.0                 |
| <b>1171</b> | 7.1           | 0.590            | 0.00        | 2.20           | 0.078     | 26.0                |
| <b>1172</b> | 9.7           | 0.420            | 0.46        | 2.10           | 0.074     | 5.0                 |
| <b>1173</b> | 7.6           | 0.360            | 0.31        | 1.70           | 0.079     | 26.0                |
| <b>1175</b> | 6.5           | 0.610            | 0.00        | 2.20           | 0.095     | 48.0                |
| <b>1176</b> | 6.5           | 0.880            | 0.03        | 5.60           | 0.079     | 23.0                |
| <b>1177</b> | 7.1           | 0.660            | 0.00        | 2.40           | 0.052     | 6.0                 |
| <b>1178</b> | 5.6           | 0.915            | 0.00        | 2.10           | 0.041     | 17.0                |
| <b>1179</b> | 8.2           | 0.350            | 0.33        | 2.40           | 0.076     | 11.0                |
| <b>1181</b> | 9.8           | 0.390            | 0.43        | 1.65           | 0.068     | 5.0                 |
| <b>1182</b> | 10.2          | 0.400            | 0.40        | 2.50           | 0.068     | 41.0                |
| <b>1183</b> | 6.8           | 0.660            | 0.07        | 1.60           | 0.070     | 16.0                |
| <b>1184</b> | 6.7           | 0.640            | 0.23        | 2.10           | 0.080     | 11.0                |
| <b>1185</b> | 7.0           | 0.430            | 0.30        | 2.00           | 0.085     | 6.0                 |
| <b>1186</b> | 6.6           | 0.800            | 0.03        | 7.80           | 0.079     | 6.0                 |
| <b>1189</b> | 8.8           | 0.955            | 0.05        | 1.80           | 0.075     | 5.0                 |
| <b>1190</b> | 9.1           | 0.400            | 0.57        | 4.60           | 0.080     | 6.0                 |
| <b>1191</b> | 6.5           | 0.885            | 0.00        | 2.30           | 0.166     | 6.0                 |
| <b>1192</b> | 7.2           | 0.250            | 0.37        | 2.50           | 0.063     | 11.0                |
| <b>1193</b> | 6.4           | 0.885            | 0.00        | 2.30           | 0.166     | 6.0                 |
| <b>1194</b> | 7.0           | 0.745            | 0.12        | 1.80           | 0.114     | 15.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1195</b> | 6.2           | 0.430            | 0.22        | 1.80           | 0.078     | 21.0                |
| <b>1196</b> | 7.9           | 0.580            | 0.23        | 2.30           | 0.076     | 23.0                |
| <b>1197</b> | 7.7           | 0.570            | 0.21        | 1.50           | 0.069     | 4.0                 |
| <b>1198</b> | 7.7           | 0.260            | 0.26        | 2.00           | 0.052     | 19.0                |
| <b>1201</b> | 7.9           | 0.340            | 0.36        | 1.90           | 0.065     | 5.0                 |
| <b>1202</b> | 8.6           | 0.420            | 0.39        | 1.80           | 0.068     | 6.0                 |
| <b>1203</b> | 9.9           | 0.740            | 0.19        | 5.80           | 0.111     | 33.0                |
| <b>1204</b> | 7.2           | 0.360            | 0.46        | 2.10           | 0.074     | 24.0                |
| <b>1207</b> | 9.9           | 0.720            | 0.55        | 1.70           | 0.136     | 24.0                |
| <b>1209</b> | 6.2           | 0.390            | 0.43        | 2.00           | 0.071     | 14.0                |
| <b>1210</b> | 6.8           | 0.650            | 0.02        | 2.10           | 0.078     | 8.0                 |
| <b>1211</b> | 6.6           | 0.440            | 0.15        | 2.10           | 0.076     | 22.0                |
| <b>1213</b> | 9.6           | 0.380            | 0.42        | 1.90           | 0.071     | 5.0                 |
| <b>1214</b> | 10.2          | 0.330            | 0.46        | 1.90           | 0.081     | 6.0                 |
| <b>1215</b> | 8.8           | 0.270            | 0.46        | 2.10           | 0.095     | 20.0                |
| <b>1216</b> | 7.9           | 0.570            | 0.31        | 2.00           | 0.079     | 10.0                |
| <b>1217</b> | 8.2           | 0.340            | 0.37        | 1.90           | 0.057     | 43.0                |
| <b>1218</b> | 8.2           | 0.400            | 0.31        | 1.90           | 0.082     | 8.0                 |
| <b>1219</b> | 9.0           | 0.390            | 0.40        | 1.30           | 0.044     | 25.0                |
| <b>1220</b> | 10.9          | 0.320            | 0.52        | 1.80           | 0.132     | 17.0                |
| <b>1222</b> | 8.1           | 0.530            | 0.22        | 2.20           | 0.078     | 33.0                |
| <b>1223</b> | 10.5          | 0.360            | 0.47        | 2.20           | 0.074     | 9.0                 |
| <b>1224</b> | 12.6          | 0.390            | 0.49        | 2.50           | 0.080     | 8.0                 |
| <b>1225</b> | 9.2           | 0.460            | 0.23        | 2.60           | 0.091     | 18.0                |
| <b>1226</b> | 7.5           | 0.580            | 0.03        | 4.10           | 0.080     | 27.0                |
| <b>1227</b> | 9.0           | 0.580            | 0.25        | 2.00           | 0.104     | 8.0                 |
| <b>1228</b> | 5.1           | 0.420            | 0.00        | 1.80           | 0.044     | 18.0                |
| <b>1229</b> | 7.6           | 0.430            | 0.29        | 2.10           | 0.075     | 19.0                |
| <b>1230</b> | 7.7           | 0.180            | 0.34        | 2.70           | 0.066     | 15.0                |
| <b>1231</b> | 7.8           | 0.815            | 0.01        | 2.60           | 0.074     | 48.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1233</b> | 10.2          | 0.230            | 0.37        | 2.20           | 0.057     | 14.0                |
| <b>1234</b> | 7.1           | 0.750            | 0.01        | 2.20           | 0.059     | 11.0                |
| <b>1235</b> | 6.0           | 0.330            | 0.32        | 12.90          | 0.054     | 6.0                 |
| <b>1236</b> | 7.8           | 0.550            | 0.00        | 1.70           | 0.070     | 7.0                 |
| <b>1238</b> | 8.1           | 0.730            | 0.00        | 2.50           | 0.081     | 12.0                |
| <b>1239</b> | 6.5           | 0.670            | 0.00        | 4.30           | 0.057     | 11.0                |
| <b>1240</b> | 7.5           | 0.610            | 0.20        | 1.70           | 0.076     | 36.0                |
| <b>1241</b> | 9.8           | 0.370            | 0.39        | 2.50           | 0.079     | 28.0                |
| <b>1242</b> | 9.0           | 0.400            | 0.41        | 2.00           | 0.058     | 15.0                |
| <b>1243</b> | 8.3           | 0.560            | 0.22        | 2.40           | 0.082     | 10.0                |
| <b>1244</b> | 5.9           | 0.290            | 0.25        | 13.40          | 0.067     | 72.0                |
| <b>1245</b> | 7.4           | 0.550            | 0.19        | 1.80           | 0.082     | 15.0                |
| <b>1246</b> | 7.4           | 0.740            | 0.07        | 1.70           | 0.086     | 15.0                |
| <b>1248</b> | 6.9           | 0.410            | 0.33        | 2.20           | 0.081     | 22.0                |
| <b>1249</b> | 7.1           | 0.600            | 0.01        | 2.30           | 0.079     | 24.0                |
| <b>1251</b> | 7.5           | 0.580            | 0.14        | 2.20           | 0.077     | 27.0                |
| <b>1252</b> | 7.1           | 0.720            | 0.00        | 1.80           | 0.123     | 6.0                 |
| <b>1253</b> | 7.9           | 0.660            | 0.00        | 1.40           | 0.096     | 6.0                 |
| <b>1254</b> | 7.8           | 0.700            | 0.06        | 1.90           | 0.079     | 20.0                |
| <b>1255</b> | 6.1           | 0.640            | 0.02        | 2.40           | 0.069     | 26.0                |
| <b>1256</b> | 7.5           | 0.590            | 0.22        | 1.80           | 0.082     | 43.0                |
| <b>1257</b> | 7.0           | 0.580            | 0.28        | 4.80           | 0.085     | 12.0                |
| <b>1258</b> | 6.8           | 0.640            | 0.00        | 2.70           | 0.123     | 15.0                |
| <b>1260</b> | 8.6           | 0.635            | 0.68        | 1.80           | 0.403     | 19.0                |
| <b>1261</b> | 6.3           | 1.020            | 0.00        | 2.00           | 0.083     | 17.0                |
| <b>1262</b> | 9.8           | 0.450            | 0.38        | 2.50           | 0.081     | 34.0                |
| <b>1263</b> | 8.2           | 0.780            | 0.00        | 2.20           | 0.089     | 13.0                |
| <b>1264</b> | 8.5           | 0.370            | 0.32        | 1.80           | 0.066     | 26.0                |
| <b>1265</b> | 7.2           | 0.570            | 0.05        | 2.30           | 0.081     | 16.0                |
| <b>1267</b> | 10.4          | 0.430            | 0.50        | 2.30           | 0.068     | 13.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1268</b> | 6.9           | 0.410            | 0.31        | 2.00           | 0.079     | 21.0                |
| <b>1269</b> | 5.5           | 0.490            | 0.03        | 1.80           | 0.044     | 28.0                |
| <b>1270</b> | 5.0           | 0.380            | 0.01        | 1.60           | 0.048     | 26.0                |
| <b>1271</b> | 7.3           | 0.440            | 0.20        | 1.60           | 0.049     | 24.0                |
| <b>1272</b> | 5.9           | 0.460            | 0.00        | 1.90           | 0.077     | 25.0                |
| <b>1273</b> | 7.5           | 0.580            | 0.20        | 2.00           | 0.073     | 34.0                |
| <b>1274</b> | 7.8           | 0.580            | 0.13        | 2.10           | 0.102     | 17.0                |
| <b>1275</b> | 8.0           | 0.715            | 0.22        | 2.30           | 0.075     | 13.0                |
| <b>1276</b> | 8.5           | 0.400            | 0.40        | 6.30           | 0.050     | 3.0                 |
| <b>1277</b> | 7.0           | 0.690            | 0.00        | 1.90           | 0.114     | 3.0                 |
| <b>1279</b> | 9.8           | 0.300            | 0.39        | 1.70           | 0.062     | 3.0                 |
| <b>1280</b> | 7.1           | 0.460            | 0.20        | 1.90           | 0.077     | 28.0                |
| <b>1282</b> | 7.9           | 0.765            | 0.00        | 2.00           | 0.084     | 9.0                 |
| <b>1283</b> | 8.7           | 0.630            | 0.28        | 2.70           | 0.096     | 17.0                |
| <b>1284</b> | 7.0           | 0.420            | 0.19        | 2.30           | 0.071     | 18.0                |
| <b>1285</b> | 11.3          | 0.370            | 0.50        | 1.80           | 0.090     | 20.0                |
| <b>1286</b> | 7.1           | 0.160            | 0.44        | 2.50           | 0.068     | 17.0                |
| <b>1287</b> | 8.0           | 0.600            | 0.08        | 2.60           | 0.056     | 3.0                 |
| <b>1288</b> | 7.0           | 0.600            | 0.30        | 4.50           | 0.068     | 20.0                |
| <b>1290</b> | 7.6           | 0.740            | 0.00        | 1.90           | 0.100     | 6.0                 |
| <b>1291</b> | 8.2           | 0.635            | 0.10        | 2.10           | 0.073     | 25.0                |
| <b>1292</b> | 5.9           | 0.395            | 0.13        | 2.40           | 0.056     | 14.0                |
| <b>1293</b> | 7.5           | 0.755            | 0.00        | 1.90           | 0.084     | 6.0                 |
| <b>1295</b> | 6.6           | 0.630            | 0.00        | 4.30           | 0.093     | 51.0                |
| <b>1297</b> | 7.2           | 0.530            | 0.14        | 2.10           | 0.064     | 15.0                |
| <b>1298</b> | 5.7           | 0.600            | 0.00        | 1.40           | 0.063     | 11.0                |
| <b>1299</b> | 7.6           | 1.580            | 0.00        | 2.10           | 0.137     | 5.0                 |
| <b>1300</b> | 5.2           | 0.645            | 0.00        | 2.15           | 0.080     | 15.0                |
| <b>1301</b> | 6.7           | 0.860            | 0.07        | 2.00           | 0.100     | 20.0                |
| <b>1302</b> | 9.1           | 0.370            | 0.32        | 2.10           | 0.064     | 4.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1303</b> | 8.0           | 0.280            | 0.44        | 1.80           | 0.081     | 28.0                |
| <b>1304</b> | 7.6           | 0.790            | 0.21        | 2.30           | 0.087     | 21.0                |
| <b>1305</b> | 7.5           | 0.610            | 0.26        | 1.90           | 0.073     | 24.0                |
| <b>1306</b> | 9.7           | 0.690            | 0.32        | 2.50           | 0.088     | 22.0                |
| <b>1307</b> | 6.8           | 0.680            | 0.09        | 3.90           | 0.068     | 15.0                |
| <b>1309</b> | 7.0           | 0.620            | 0.10        | 1.40           | 0.071     | 27.0                |
| <b>1311</b> | 6.5           | 0.510            | 0.15        | 3.00           | 0.064     | 12.0                |
| <b>1312</b> | 8.0           | 1.180            | 0.21        | 1.90           | 0.083     | 14.0                |
| <b>1313</b> | 7.0           | 0.360            | 0.21        | 2.30           | 0.086     | 20.0                |
| <b>1314</b> | 7.0           | 0.360            | 0.21        | 2.40           | 0.086     | 24.0                |
| <b>1315</b> | 7.5           | 0.630            | 0.27        | 2.00           | 0.083     | 17.0                |
| <b>1316</b> | 5.4           | 0.740            | 0.00        | 1.20           | 0.041     | 16.0                |
| <b>1317</b> | 9.9           | 0.440            | 0.46        | 2.20           | 0.091     | 10.0                |
| <b>1319</b> | 9.1           | 0.760            | 0.68        | 1.70           | 0.414     | 18.0                |
| <b>1320</b> | 9.7           | 0.660            | 0.34        | 2.60           | 0.094     | 12.0                |
| <b>1321</b> | 5.0           | 0.740            | 0.00        | 1.20           | 0.041     | 16.0                |
| <b>1322</b> | 9.1           | 0.340            | 0.42        | 1.80           | 0.058     | 9.0                 |
| <b>1323</b> | 9.1           | 0.360            | 0.39        | 1.80           | 0.060     | 21.0                |
| <b>1324</b> | 6.7           | 0.460            | 0.24        | 1.70           | 0.077     | 18.0                |
| <b>1328</b> | 6.5           | 0.520            | 0.11        | 1.80           | 0.073     | 13.0                |
| <b>1329</b> | 7.4           | 0.600            | 0.26        | 2.10           | 0.083     | 17.0                |
| <b>1331</b> | 7.8           | 0.870            | 0.26        | 3.80           | 0.107     | 31.0                |
| <b>1332</b> | 8.4           | 0.390            | 0.10        | 1.70           | 0.075     | 6.0                 |
| <b>1333</b> | 9.1           | 0.775            | 0.22        | 2.20           | 0.079     | 12.0                |
| <b>1334</b> | 7.2           | 0.835            | 0.00        | 2.00           | 0.166     | 4.0                 |
| <b>1335</b> | 6.6           | 0.580            | 0.02        | 2.40           | 0.069     | 19.0                |
| <b>1336</b> | 6.0           | 0.500            | 0.00        | 1.40           | 0.057     | 15.0                |
| <b>1339</b> | 7.5           | 0.510            | 0.02        | 1.70           | 0.084     | 13.0                |
| <b>1342</b> | 7.6           | 0.540            | 0.02        | 1.70           | 0.085     | 17.0                |
| <b>1344</b> | 11.5          | 0.420            | 0.48        | 2.60           | 0.077     | 8.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1345</b> | 8.2           | 0.440            | 0.24        | 2.30           | 0.063     | 10.0                |
| <b>1346</b> | 6.1           | 0.590            | 0.01        | 2.10           | 0.056     | 5.0                 |
| <b>1347</b> | 7.2           | 0.655            | 0.03        | 1.80           | 0.078     | 7.0                 |
| <b>1349</b> | 6.9           | 0.570            | 0.00        | 2.80           | 0.081     | 21.0                |
| <b>1350</b> | 9.0           | 0.600            | 0.29        | 2.00           | 0.069     | 32.0                |
| <b>1351</b> | 7.2           | 0.620            | 0.01        | 2.30           | 0.065     | 8.0                 |
| <b>1352</b> | 7.6           | 0.645            | 0.03        | 1.90           | 0.086     | 14.0                |
| <b>1354</b> | 7.2           | 0.580            | 0.03        | 2.30           | 0.077     | 7.0                 |
| <b>1355</b> | 6.1           | 0.320            | 0.25        | 1.80           | 0.086     | 5.0                 |
| <b>1356</b> | 6.1           | 0.340            | 0.25        | 1.80           | 0.084     | 4.0                 |
| <b>1357</b> | 7.3           | 0.430            | 0.24        | 2.50           | 0.078     | 27.0                |
| <b>1358</b> | 7.4           | 0.640            | 0.17        | 5.40           | 0.168     | 52.0                |
| <b>1359</b> | 11.6          | 0.475            | 0.40        | 1.40           | 0.091     | 6.0                 |
| <b>1360</b> | 9.2           | 0.540            | 0.31        | 2.30           | 0.112     | 11.0                |
| <b>1361</b> | 8.3           | 0.850            | 0.14        | 2.50           | 0.093     | 13.0                |
| <b>1363</b> | 8.0           | 0.830            | 0.27        | 2.00           | 0.080     | 11.0                |
| <b>1364</b> | 7.2           | 0.605            | 0.02        | 1.90           | 0.096     | 10.0                |
| <b>1365</b> | 7.8           | 0.500            | 0.09        | 2.20           | 0.115     | 10.0                |
| <b>1366</b> | 7.3           | 0.740            | 0.08        | 1.70           | 0.094     | 10.0                |
| <b>1367</b> | 6.9           | 0.540            | 0.30        | 2.20           | 0.088     | 9.0                 |
| <b>1368</b> | 8.0           | 0.770            | 0.32        | 2.10           | 0.079     | 16.0                |
| <b>1369</b> | 6.6           | 0.610            | 0.00        | 1.60           | 0.069     | 4.0                 |
| <b>1370</b> | 8.7           | 0.780            | 0.51        | 1.70           | 0.415     | 12.0                |
| <b>1371</b> | 7.5           | 0.580            | 0.56        | 3.10           | 0.153     | 5.0                 |
| <b>1373</b> | 7.7           | 0.750            | 0.27        | 3.80           | 0.110     | 34.0                |
| <b>1374</b> | 6.8           | 0.815            | 0.00        | 1.20           | 0.267     | 16.0                |
| <b>1375</b> | 7.2           | 0.560            | 0.26        | 2.00           | 0.083     | 13.0                |
| <b>1376</b> | 8.2           | 0.885            | 0.20        | 1.40           | 0.086     | 7.0                 |
| <b>1377</b> | 5.2           | 0.490            | 0.26        | 2.30           | 0.090     | 23.0                |
| <b>1378</b> | 7.2           | 0.450            | 0.15        | 2.00           | 0.078     | 10.0                |



|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1379</b> | 7.5           | 0.570            | 0.02        | 2.60           | 0.077     | 11.0                |
| <b>1381</b> | 6.8           | 0.830            | 0.09        | 1.80           | 0.074     | 4.0                 |
| <b>1382</b> | 8.0           | 0.600            | 0.22        | 2.10           | 0.080     | 25.0                |
| <b>1384</b> | 7.1           | 0.755            | 0.15        | 1.80           | 0.107     | 20.0                |
| <b>1385</b> | 8.0           | 0.810            | 0.25        | 3.40           | 0.076     | 34.0                |
| <b>1386</b> | 7.4           | 0.640            | 0.07        | 1.80           | 0.100     | 8.0                 |
| <b>1388</b> | 6.6           | 0.640            | 0.31        | 6.10           | 0.083     | 7.0                 |
| <b>1389</b> | 6.7           | 0.480            | 0.02        | 2.20           | 0.080     | 36.0                |
| <b>1390</b> | 6.0           | 0.490            | 0.00        | 2.30           | 0.068     | 15.0                |
| <b>1391</b> | 8.0           | 0.640            | 0.22        | 2.40           | 0.094     | 5.0                 |
| <b>1392</b> | 7.1           | 0.620            | 0.06        | 1.30           | 0.070     | 5.0                 |
| <b>1393</b> | 8.0           | 0.520            | 0.25        | 2.00           | 0.078     | 19.0                |
| <b>1394</b> | 6.4           | 0.570            | 0.14        | 3.90           | 0.070     | 27.0                |
| <b>1395</b> | 8.6           | 0.685            | 0.10        | 1.60           | 0.092     | 3.0                 |
| <b>1396</b> | 8.7           | 0.675            | 0.10        | 1.60           | 0.090     | 4.0                 |
| <b>1397</b> | 7.3           | 0.590            | 0.26        | 2.00           | 0.080     | 17.0                |
| <b>1398</b> | 7.0           | 0.600            | 0.12        | 2.20           | 0.083     | 13.0                |
| <b>1399</b> | 7.2           | 0.670            | 0.00        | 2.20           | 0.068     | 10.0                |
| <b>1400</b> | 7.9           | 0.690            | 0.21        | 2.10           | 0.080     | 33.0                |
| <b>1402</b> | 7.6           | 0.300            | 0.42        | 2.00           | 0.052     | 6.0                 |
| <b>1403</b> | 7.2           | 0.330            | 0.33        | 1.70           | 0.061     | 3.0                 |
| <b>1404</b> | 8.0           | 0.500            | 0.39        | 2.60           | 0.082     | 12.0                |
| <b>1405</b> | 7.7           | 0.280            | 0.30        | 2.00           | 0.062     | 18.0                |
| <b>1406</b> | 8.2           | 0.240            | 0.34        | 5.10           | 0.062     | 8.0                 |
| <b>1407</b> | 6.0           | 0.510            | 0.00        | 2.10           | 0.064     | 40.0                |
| <b>1408</b> | 8.1           | 0.290            | 0.36        | 2.20           | 0.048     | 35.0                |
| <b>1410</b> | 6.6           | 0.960            | 0.00        | 1.80           | 0.082     | 5.0                 |
| <b>1411</b> | 6.4           | 0.470            | 0.40        | 2.40           | 0.071     | 8.0                 |
| <b>1413</b> | 9.9           | 0.570            | 0.25        | 2.00           | 0.104     | 12.0                |
| <b>1414</b> | 10.0          | 0.320            | 0.59        | 2.20           | 0.077     | 3.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1415</b> | 6.2           | 0.580            | 0.00        | 1.60           | 0.065     | 8.0                 |
| <b>1417</b> | 7.3           | 0.340            | 0.33        | 2.50           | 0.064     | 21.0                |
| <b>1418</b> | 7.8           | 0.530            | 0.01        | 1.60           | 0.077     | 3.0                 |
| <b>1419</b> | 7.7           | 0.640            | 0.21        | 2.20           | 0.077     | 32.0                |
| <b>1421</b> | 7.5           | 0.400            | 0.18        | 1.60           | 0.079     | 24.0                |
| <b>1422</b> | 7.0           | 0.540            | 0.00        | 2.10           | 0.079     | 39.0                |
| <b>1423</b> | 6.4           | 0.530            | 0.09        | 3.90           | 0.123     | 14.0                |
| <b>1424</b> | 8.3           | 0.260            | 0.37        | 1.40           | 0.076     | 8.0                 |
| <b>1426</b> | 7.7           | 0.230            | 0.37        | 1.80           | 0.046     | 23.0                |
| <b>1427</b> | 7.6           | 0.410            | 0.33        | 2.50           | 0.078     | 6.0                 |
| <b>1428</b> | 7.8           | 0.640            | 0.00        | 1.90           | 0.072     | 27.0                |
| <b>1429</b> | 7.9           | 0.180            | 0.40        | 2.20           | 0.049     | 38.0                |
| <b>1430</b> | 7.4           | 0.410            | 0.24        | 1.80           | 0.066     | 18.0                |
| <b>1431</b> | 7.6           | 0.430            | 0.31        | 2.10           | 0.069     | 13.0                |
| <b>1432</b> | 5.9           | 0.440            | 0.00        | 1.60           | 0.042     | 3.0                 |
| <b>1433</b> | 6.1           | 0.400            | 0.16        | 1.80           | 0.069     | 11.0                |
| <b>1434</b> | 10.2          | 0.540            | 0.37        | 15.40          | 0.214     | 55.0                |
| <b>1436</b> | 10.0          | 0.380            | 0.38        | 1.60           | 0.169     | 27.0                |
| <b>1437</b> | 6.8           | 0.915            | 0.29        | 4.80           | 0.070     | 15.0                |
| <b>1438</b> | 7.0           | 0.590            | 0.00        | 1.70           | 0.052     | 3.0                 |
| <b>1439</b> | 7.3           | 0.670            | 0.02        | 2.20           | 0.072     | 31.0                |
| <b>1440</b> | 7.2           | 0.370            | 0.32        | 2.00           | 0.062     | 15.0                |
| <b>1441</b> | 7.4           | 0.785            | 0.19        | 5.20           | 0.094     | 19.0                |
| <b>1442</b> | 6.9           | 0.630            | 0.02        | 1.90           | 0.078     | 18.0                |
| <b>1443</b> | 6.9           | 0.580            | 0.20        | 1.75           | 0.058     | 8.0                 |
| <b>1444</b> | 7.3           | 0.670            | 0.02        | 2.20           | 0.072     | 31.0                |
| <b>1445</b> | 7.4           | 0.785            | 0.19        | 5.20           | 0.094     | 19.0                |
| <b>1447</b> | 6.8           | 0.670            | 0.00        | 1.90           | 0.080     | 22.0                |
| <b>1448</b> | 6.9           | 0.580            | 0.01        | 1.90           | 0.080     | 40.0                |
| <b>1449</b> | 7.2           | 0.380            | 0.31        | 2.00           | 0.056     | 15.0                |

|      | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------|---------------|------------------|-------------|----------------|-----------|---------------------|
| 1451 | 7.8           | 0.320            | 0.44        | 2.70           | 0.104     | 8.0                 |
| 1452 | 6.6           | 0.580            | 0.02        | 2.00           | 0.062     | 37.0                |
| 1453 | 7.6           | 0.490            | 0.33        | 1.90           | 0.074     | 27.0                |
| 1454 | 11.7          | 0.450            | 0.63        | 2.20           | 0.073     | 7.0                 |
| 1455 | 6.5           | 0.900            | 0.00        | 1.60           | 0.052     | 9.0                 |
| 1456 | 6.0           | 0.540            | 0.06        | 1.80           | 0.050     | 38.0                |
| 1458 | 8.4           | 0.290            | 0.40        | 1.70           | 0.067     | 8.0                 |
| 1459 | 7.9           | 0.200            | 0.35        | 1.70           | 0.054     | 7.0                 |
| 1460 | 6.4           | 0.420            | 0.09        | 2.30           | 0.054     | 34.0                |
| 1461 | 6.2           | 0.785            | 0.00        | 2.10           | 0.060     | 6.0                 |
| 1462 | 6.8           | 0.640            | 0.03        | 2.30           | 0.075     | 14.0                |
| 1463 | 6.9           | 0.630            | 0.01        | 2.40           | 0.076     | 14.0                |
| 1464 | 6.8           | 0.590            | 0.10        | 1.70           | 0.063     | 34.0                |
| 1466 | 7.3           | 0.480            | 0.32        | 2.10           | 0.062     | 31.0                |
| 1467 | 6.7           | 1.040            | 0.08        | 2.30           | 0.067     | 19.0                |
| 1469 | 7.3           | 0.980            | 0.05        | 2.10           | 0.061     | 20.0                |
| 1470 | 10.0          | 0.690            | 0.11        | 1.40           | 0.084     | 8.0                 |
| 1471 | 6.7           | 0.700            | 0.08        | 3.75           | 0.067     | 8.0                 |
| 1472 | 7.6           | 0.350            | 0.60        | 2.60           | 0.073     | 23.0                |
| 1473 | 6.1           | 0.600            | 0.08        | 1.80           | 0.071     | 14.0                |
| 1474 | 9.9           | 0.500            | 0.50        | 13.80          | 0.205     | 48.0                |
| 1475 | 5.3           | 0.470            | 0.11        | 2.20           | 0.048     | 16.0                |
| 1477 | 5.3           | 0.470            | 0.11        | 2.20           | 0.048     | 16.0                |
| 1478 | 7.1           | 0.875            | 0.05        | 5.70           | 0.082     | 3.0                 |
| 1479 | 8.2           | 0.280            | 0.60        | 3.00           | 0.104     | 10.0                |
| 1480 | 5.6           | 0.620            | 0.03        | 1.50           | 0.080     | 6.0                 |
| 1482 | 7.2           | 0.580            | 0.54        | 2.10           | 0.114     | 3.0                 |
| 1483 | 8.1           | 0.330            | 0.44        | 1.50           | 0.042     | 6.0                 |
| 1484 | 6.8           | 0.910            | 0.06        | 2.00           | 0.060     | 4.0                 |
| 1485 | 7.0           | 0.655            | 0.16        | 2.10           | 0.074     | 8.0                 |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1486</b> | 6.8           | 0.680            | 0.21        | 2.10           | 0.070     | 9.0                 |
| <b>1487</b> | 6.0           | 0.640            | 0.05        | 1.90           | 0.066     | 9.0                 |
| <b>1488</b> | 5.6           | 0.540            | 0.04        | 1.70           | 0.049     | 5.0                 |
| <b>1489</b> | 6.2           | 0.570            | 0.10        | 2.10           | 0.048     | 4.0                 |
| <b>1490</b> | 7.1           | 0.220            | 0.49        | 1.80           | 0.039     | 8.0                 |
| <b>1492</b> | 6.2           | 0.650            | 0.06        | 1.60           | 0.050     | 6.0                 |
| <b>1493</b> | 7.7           | 0.540            | 0.26        | 1.90           | 0.089     | 23.0                |
| <b>1494</b> | 6.4           | 0.310            | 0.09        | 1.40           | 0.066     | 15.0                |
| <b>1495</b> | 7.0           | 0.430            | 0.02        | 1.90           | 0.080     | 15.0                |
| <b>1497</b> | 6.9           | 0.740            | 0.03        | 2.30           | 0.054     | 7.0                 |
| <b>1498</b> | 6.6           | 0.895            | 0.04        | 2.30           | 0.068     | 7.0                 |
| <b>1500</b> | 7.5           | 0.725            | 0.04        | 1.50           | 0.076     | 8.0                 |
| <b>1501</b> | 7.8           | 0.820            | 0.29        | 4.30           | 0.083     | 21.0                |
| <b>1502</b> | 7.3           | 0.585            | 0.18        | 2.40           | 0.078     | 15.0                |
| <b>1503</b> | 6.2           | 0.440            | 0.39        | 2.50           | 0.077     | 6.0                 |
| <b>1504</b> | 7.5           | 0.380            | 0.57        | 2.30           | 0.106     | 5.0                 |
| <b>1505</b> | 6.7           | 0.760            | 0.02        | 1.80           | 0.078     | 6.0                 |
| <b>1506</b> | 6.8           | 0.810            | 0.05        | 2.00           | 0.070     | 6.0                 |
| <b>1508</b> | 7.1           | 0.270            | 0.60        | 2.10           | 0.074     | 17.0                |
| <b>1509</b> | 7.9           | 0.180            | 0.40        | 1.80           | 0.062     | 7.0                 |
| <b>1510</b> | 6.4           | 0.360            | 0.21        | 2.20           | 0.047     | 26.0                |
| <b>1511</b> | 7.1           | 0.690            | 0.04        | 2.10           | 0.068     | 19.0                |
| <b>1512</b> | 6.4           | 0.790            | 0.04        | 2.20           | 0.061     | 11.0                |
| <b>1513</b> | 6.4           | 0.560            | 0.15        | 1.80           | 0.078     | 17.0                |
| <b>1514</b> | 6.9           | 0.840            | 0.21        | 4.10           | 0.074     | 16.0                |
| <b>1515</b> | 6.9           | 0.840            | 0.21        | 4.10           | 0.074     | 16.0                |
| <b>1516</b> | 6.1           | 0.320            | 0.25        | 2.30           | 0.071     | 23.0                |
| <b>1517</b> | 6.5           | 0.530            | 0.06        | 2.00           | 0.063     | 29.0                |
| <b>1518</b> | 7.4           | 0.470            | 0.46        | 2.20           | 0.114     | 7.0                 |
| <b>1519</b> | 6.6           | 0.700            | 0.08        | 2.60           | 0.106     | 14.0                |

|      | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|------|---------------|------------------|-------------|----------------|-----------|---------------------|
| 1521 | 6.9           | 0.480            | 0.20        | 1.90           | 0.082     | 9.0                 |
| 1523 | 6.8           | 0.480            | 0.25        | 2.00           | 0.076     | 29.0                |
| 1524 | 6.0           | 0.420            | 0.19        | 2.00           | 0.075     | 22.0                |
| 1525 | 6.7           | 0.480            | 0.08        | 2.10           | 0.064     | 18.0                |
| 1526 | 6.8           | 0.470            | 0.08        | 2.20           | 0.064     | 18.0                |
| 1527 | 7.1           | 0.530            | 0.07        | 1.70           | 0.071     | 15.0                |
| 1528 | 7.9           | 0.290            | 0.49        | 2.20           | 0.096     | 21.0                |
| 1529 | 7.1           | 0.690            | 0.08        | 2.10           | 0.063     | 42.0                |
| 1530 | 6.6           | 0.440            | 0.09        | 2.20           | 0.063     | 9.0                 |
| 1531 | 6.1           | 0.705            | 0.10        | 2.80           | 0.081     | 13.0                |
| 1532 | 7.2           | 0.530            | 0.13        | 2.00           | 0.058     | 18.0                |
| 1533 | 8.0           | 0.390            | 0.30        | 1.90           | 0.074     | 32.0                |
| 1534 | 6.6           | 0.560            | 0.14        | 2.40           | 0.064     | 13.0                |
| 1535 | 7.0           | 0.550            | 0.13        | 2.20           | 0.075     | 15.0                |
| 1536 | 6.1           | 0.530            | 0.08        | 1.90           | 0.077     | 24.0                |
| 1537 | 5.4           | 0.580            | 0.08        | 1.90           | 0.059     | 20.0                |
| 1538 | 6.2           | 0.640            | 0.09        | 2.50           | 0.081     | 15.0                |
| 1539 | 7.2           | 0.390            | 0.32        | 1.80           | 0.065     | 34.0                |
| 1540 | 6.2           | 0.520            | 0.08        | 4.40           | 0.071     | 11.0                |
| 1541 | 7.4           | 0.250            | 0.29        | 2.20           | 0.054     | 19.0                |
| 1542 | 6.7           | 0.855            | 0.02        | 1.90           | 0.064     | 29.0                |
| 1543 | 11.1          | 0.440            | 0.42        | 2.20           | 0.064     | 14.0                |
| 1544 | 8.4           | 0.370            | 0.43        | 2.30           | 0.063     | 12.0                |
| 1545 | 6.5           | 0.630            | 0.33        | 1.80           | 0.059     | 16.0                |
| 1546 | 7.0           | 0.570            | 0.02        | 2.00           | 0.072     | 17.0                |
| 1547 | 6.3           | 0.600            | 0.10        | 1.60           | 0.048     | 12.0                |
| 1548 | 11.2          | 0.400            | 0.50        | 2.00           | 0.099     | 19.0                |
| 1549 | 7.4           | 0.360            | 0.30        | 1.80           | 0.074     | 17.0                |
| 1550 | 7.1           | 0.680            | 0.00        | 2.30           | 0.087     | 17.0                |
| 1551 | 7.1           | 0.670            | 0.00        | 2.30           | 0.083     | 18.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1552</b> | 6.3           | 0.680            | 0.01        | 3.70           | 0.103     | 32.0                |
| <b>1553</b> | 7.3           | 0.735            | 0.00        | 2.20           | 0.080     | 18.0                |
| <b>1554</b> | 6.6           | 0.855            | 0.02        | 2.40           | 0.062     | 15.0                |
| <b>1555</b> | 7.0           | 0.560            | 0.17        | 1.70           | 0.065     | 15.0                |
| <b>1556</b> | 6.6           | 0.880            | 0.04        | 2.20           | 0.066     | 12.0                |
| <b>1558</b> | 6.9           | 0.630            | 0.33        | 6.70           | 0.235     | 66.0                |
| <b>1559</b> | 7.8           | 0.600            | 0.26        | 2.00           | 0.080     | 31.0                |
| <b>1562</b> | 7.2           | 0.695            | 0.13        | 2.00           | 0.076     | 12.0                |
| <b>1565</b> | 6.7           | 0.670            | 0.02        | 1.90           | 0.061     | 26.0                |
| <b>1566</b> | 6.7           | 0.160            | 0.64        | 2.10           | 0.059     | 24.0                |
| <b>1568</b> | 7.0           | 0.560            | 0.13        | 1.60           | 0.077     | 25.0                |
| <b>1569</b> | 6.2           | 0.510            | 0.14        | 1.90           | 0.056     | 15.0                |
| <b>1570</b> | 6.4           | 0.360            | 0.53        | 2.20           | 0.230     | 19.0                |
| <b>1571</b> | 6.4           | 0.380            | 0.14        | 2.20           | 0.038     | 15.0                |
| <b>1572</b> | 7.3           | 0.690            | 0.32        | 2.20           | 0.069     | 35.0                |
| <b>1573</b> | 6.0           | 0.580            | 0.20        | 2.40           | 0.075     | 15.0                |
| <b>1574</b> | 5.6           | 0.310            | 0.78        | 13.90          | 0.074     | 23.0                |
| <b>1575</b> | 7.5           | 0.520            | 0.40        | 2.20           | 0.060     | 12.0                |
| <b>1576</b> | 8.0           | 0.300            | 0.63        | 1.60           | 0.081     | 16.0                |
| <b>1577</b> | 6.2           | 0.700            | 0.15        | 5.10           | 0.076     | 13.0                |
| <b>1578</b> | 6.8           | 0.670            | 0.15        | 1.80           | 0.118     | 13.0                |
| <b>1579</b> | 6.2           | 0.560            | 0.09        | 1.70           | 0.053     | 24.0                |
| <b>1580</b> | 7.4           | 0.350            | 0.33        | 2.40           | 0.068     | 9.0                 |
| <b>1582</b> | 6.1           | 0.715            | 0.10        | 2.60           | 0.053     | 13.0                |
| <b>1583</b> | 6.2           | 0.460            | 0.29        | 2.10           | 0.074     | 32.0                |
| <b>1584</b> | 6.7           | 0.320            | 0.44        | 2.40           | 0.061     | 24.0                |
| <b>1585</b> | 7.2           | 0.390            | 0.44        | 2.60           | 0.066     | 22.0                |
| <b>1586</b> | 7.5           | 0.310            | 0.41        | 2.40           | 0.065     | 34.0                |
| <b>1587</b> | 5.8           | 0.610            | 0.11        | 1.80           | 0.066     | 18.0                |
| <b>1588</b> | 7.2           | 0.660            | 0.33        | 2.50           | 0.068     | 34.0                |

|             | fixed_acidity | volatile_acidity | citric_acid | residual_sugar | chlorides | free_sulfur_dioxide |
|-------------|---------------|------------------|-------------|----------------|-----------|---------------------|
| <b>1589</b> | 6.6           | 0.725            | 0.20        | 7.80           | 0.073     | 29.0                |
| <b>1590</b> | 6.3           | 0.550            | 0.15        | 1.80           | 0.077     | 26.0                |
| <b>1591</b> | 5.4           | 0.740            | 0.09        | 1.70           | 0.089     | 16.0                |
| <b>1592</b> | 6.3           | 0.510            | 0.13        | 2.30           | 0.076     | 29.0                |
| <b>1593</b> | 6.8           | 0.620            | 0.08        | 1.90           | 0.068     | 28.0                |
| <b>1594</b> | 6.2           | 0.600            | 0.08        | 2.00           | 0.090     | 32.0                |
| <b>1595</b> | 5.9           | 0.550            | 0.10        | 2.20           | 0.062     | 39.0                |
| <b>1597</b> | 5.9           | 0.645            | 0.12        | 2.00           | 0.075     | 32.0                |
| <b>1598</b> | 6.0           | 0.310            | 0.47        | 3.60           | 0.067     | 18.0                |

### (3) Các tính chất thống kê trên dữ liệu số

- Count, Mean, Standard Deviation, Minimum Value
- 25th Percentile, 50th Percentile (Median), 75th Percentile, Maximum Value

```
In [67]: description = df.describe().T
display(description)
```

|                             | count  | mean      | std       | min     | 25%     | 50%     | 75%      | r      |
|-----------------------------|--------|-----------|-----------|---------|---------|---------|----------|--------|
| <b>fixed_acidity</b>        | 1359.0 | 8.310596  | 1.736990  | 4.60000 | 7.1000  | 7.9000  | 9.20000  | 15.90  |
| <b>volatile_acidity</b>     | 1359.0 | 0.529478  | 0.183031  | 0.12000 | 0.3900  | 0.5200  | 0.64000  | 1.58   |
| <b>citric_acid</b>          | 1359.0 | 0.272333  | 0.195537  | 0.00000 | 0.0900  | 0.2600  | 0.43000  | 1.00   |
| <b>residual_sugar</b>       | 1359.0 | 2.523400  | 1.352314  | 0.90000 | 1.9000  | 2.2000  | 2.60000  | 15.50  |
| <b>chlorides</b>            | 1359.0 | 0.088124  | 0.049377  | 0.01200 | 0.0700  | 0.0790  | 0.09100  | 0.61   |
| <b>free_sulfur_dioxide</b>  | 1359.0 | 15.893304 | 10.447270 | 1.00000 | 7.0000  | 14.0000 | 21.00000 | 72.00  |
| <b>total_sulfur_dioxide</b> | 1359.0 | 46.825975 | 33.408946 | 6.00000 | 22.0000 | 38.0000 | 63.00000 | 289.00 |
| <b>density</b>              | 1359.0 | 0.996709  | 0.001869  | 0.99007 | 0.9956  | 0.9967  | 0.99782  | 1.00   |
| <b>pH</b>                   | 1359.0 | 3.309787  | 0.155036  | 2.74000 | 3.2100  | 3.3100  | 3.40000  | 4.01   |
| <b>sulphates</b>            | 1359.0 | 0.658705  | 0.170667  | 0.33000 | 0.5500  | 0.6200  | 0.73000  | 2.00   |
| <b>alcohol</b>              | 1359.0 | 10.432315 | 1.082065  | 8.40000 | 9.5000  | 10.2000 | 11.10000 | 14.90  |
| <b>quality</b>              | 1359.0 | 5.623252  | 0.823578  | 3.00000 | 5.0000  | 6.0000  | 6.00000  | 8.00   |



## 4. Phân tích dữ liệu (Analyze Data)

### 4.1 Biểu đồ tần suất (displot)

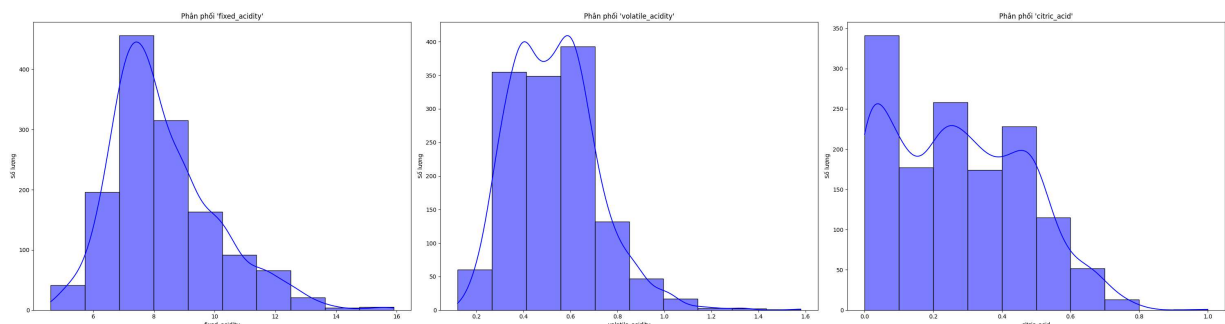
```
In [68]: fig, axes = plt.subplots(1, 3, figsize=(30, 8))

# fixed_acidity
sns.histplot(
    data=df, x="fixed_acidity",
    bins=10, kde=True, color="blue",
    ax=axes[0]
)
axes[0].set_title("Phân phối 'fixed_acidity'")
axes[0].set_xlabel("fixed_acidity")
axes[0].set_ylabel("Số lượng")

# volatile_acidity
sns.histplot(
    data=df, x="volatile_acidity",
    bins=10, kde=True, color="blue",
    ax=axes[1]
)
axes[1].set_title("Phân phối 'volatile_acidity'")
axes[1].set_xlabel("volatile_acidity")
axes[1].set_ylabel("Số lượng")

# citric_acid
sns.histplot(
    data=df, x="citric_acid",
    bins=10, kde=True, color="blue",
    ax=axes[2]
)
axes[2].set_title("Phân phối 'citric_acid'")
axes[2].set_xlabel("citric_acid")
axes[2].set_ylabel("Số lượng")

plt.tight_layout()
plt.show()
```



Qua phân tích biểu đồ cho thấy giá trị của biến `fixed_acidity` phân bố lệch trái. Trong khi đó, các biến `volatile_acidity` và `citric_acid` có nhiều đỉnh trong phân



bố.

In [69]: fig, axes = plt.subplots(1, 3, figsize=(30, 8))

```

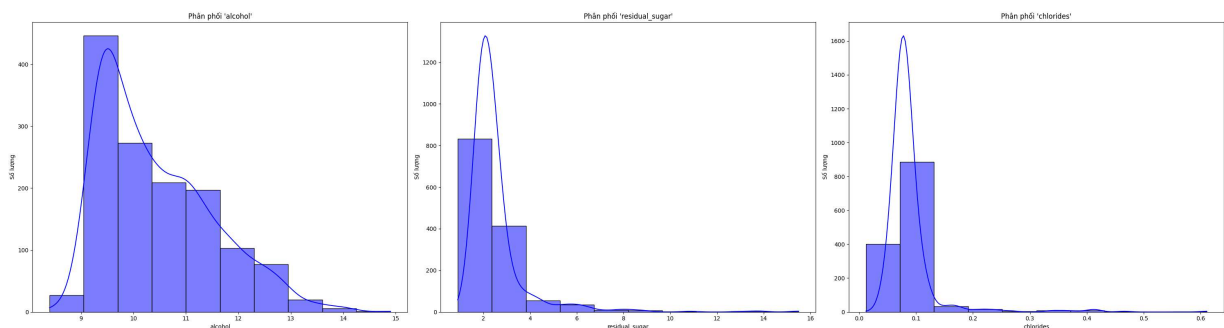
# alcohol
sns.histplot(
    data=df, x="alcohol",
    bins=10, kde=True,
    color="blue",
    ax=axes[0]
)
axes[0].set_title("Phân phối 'alcohol'")
axes[0].set_xlabel("alcohol")
axes[0].set_ylabel("Số lượng")

# residual_sugar
sns.histplot(
    data=df, x="residual_sugar",
    bins=10, kde=True,
    color="blue",
    ax=axes[1]
)
axes[1].set_title("Phân phối 'residual_sugar'")
axes[1].set_xlabel("residual_sugar")
axes[1].set_ylabel("Số lượng")

# chlorides
sns.histplot(
    data=df, x="chlorides",
    bins=10, kde=True,
    color="blue",
    ax=axes[2]
)
axes[2].set_title("Phân phối 'chlorides'")
axes[2].set_xlabel("chlorides")
axes[2].set_ylabel("Số lượng")

plt.tight_layout()
plt.show()

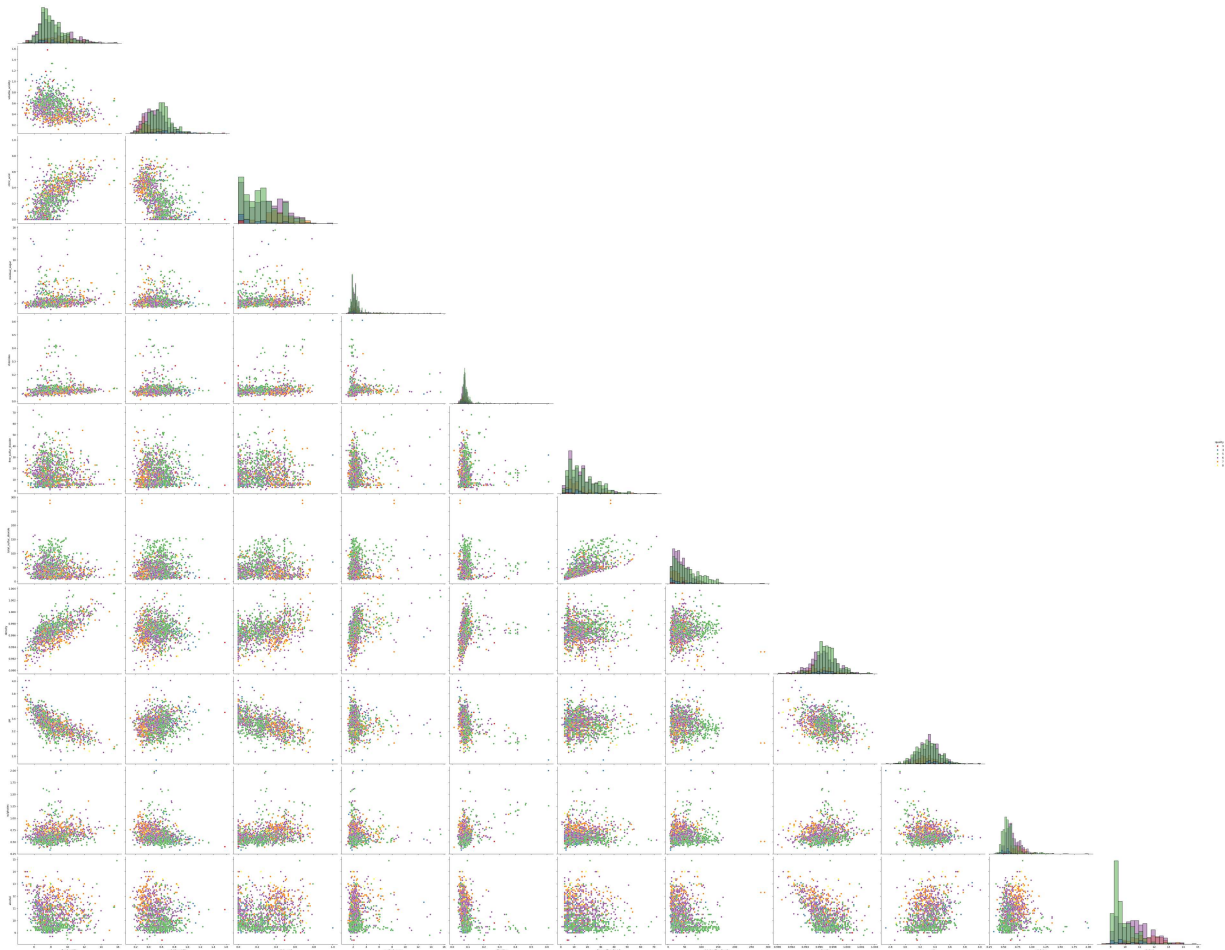
```



cả alcohol, residual\_sugar, chlorides có phân không đồng đều

## 4.2 Pairplot

```
In [70]: sns.pairplot(
    df,
    diag_kind="hist",
    hue="quality",
    height=5,
    aspect=1.2, # tăng tỷ lệ chiều ngang
    corner=True,
    palette="Set1"
);
```



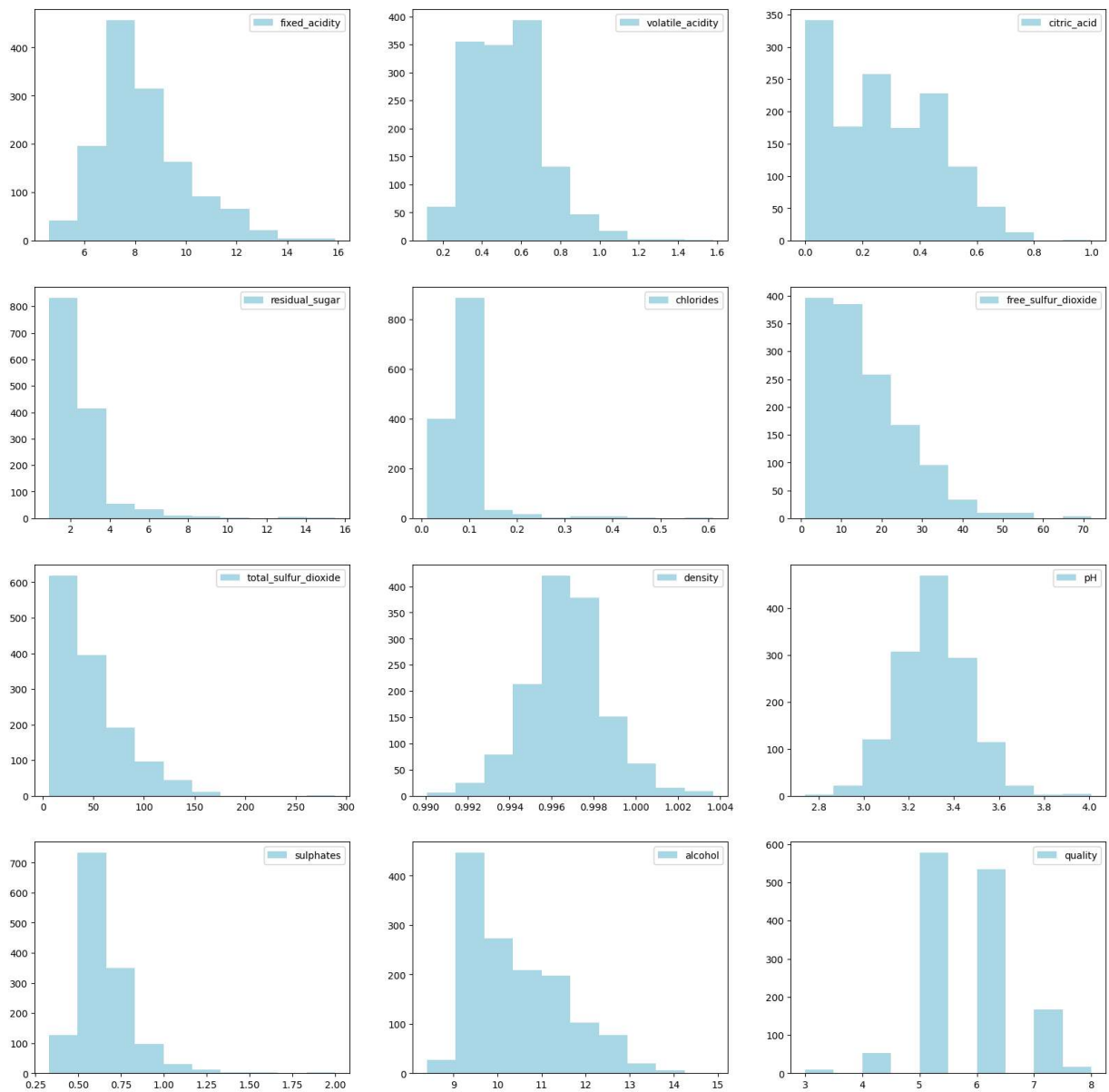
## 4.3 Histogram

```
In [71]: fig, axes = plt.subplots(nrows = 4, ncols = 3, figsize = (20, 20))

for index, column in enumerate(df.columns):
    ax = axes.flatten()[index]
    ax.hist(df[column], color = 'lightblue', label = column)
    ax.legend(loc = "best")
plt.suptitle("Histograms", size = 18)

plt.show()
```

## Histograms



- **Khoảng giá trị chủ yếu:**

- **fixed\_acidity**: 7 – 8
- **volatile\_acidity**: 0.4 – 0.7
- **citric\_acid**: 0.0 – 0.1
- **residual\_sugar**: 1 – 2.5
- **chlorides**: 0.085 – 0.15
- **free\_sulfur\_dioxide**: 0 – 15
- **total\_sulfur\_dioxide**: 0 – 30
- **density**: 0.996 – 0.998
- **pH**: 3.2 – 3.4
- **sulphates**: 0.50 – 0.75

- `alcohol`: 9 – 10
- `quality`: tập trung ở mức 5 và 6

## 4.4 Boxplot

```
In [72]: columns = [
    "fixed_acidity",
    "volatile_acidity",
    "citric_acid",
    "residual_sugar",
    "chlorides",
    "free_sulfur_dioxide",
    "total_sulfur_dioxide",
    "density",
    "pH",
    "sulphates",
    "alcohol"
]

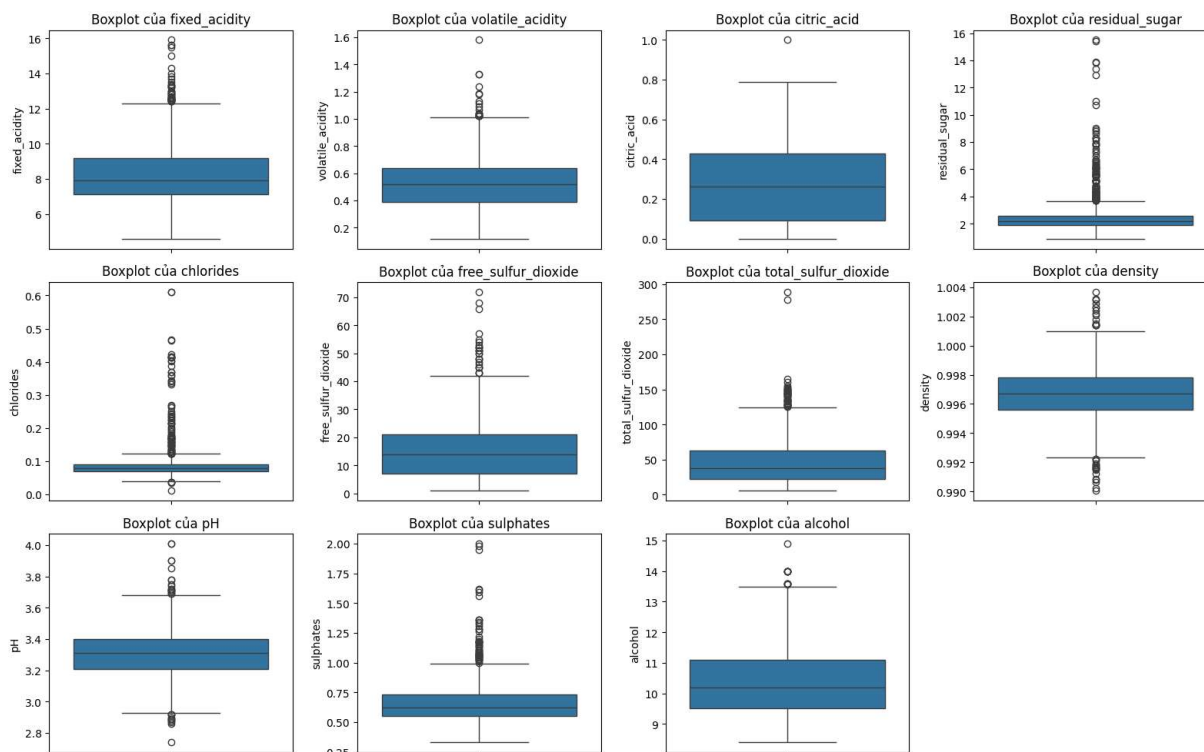
fig, axes = plt.subplots(3, 4, figsize=(16, 10))
axes = axes.flatten()

for i, col in enumerate(columns):
    sns.boxplot(y=df[col], ax=axes[i])
    axes[i].set_title(f'Boxplot của {col}')

# Ẩn subplot dư thừa nếu có
for j in range(len(columns), len(axes)):
    axes[j].set_visible(False)

plt.tight_layout()
plt.show()

# Tính Median, IQR và số lượng outlier cho từng biến
print("\nThống kê Median, IQR và số lượng Outliers:")
for col in columns:
    median = df[col].median()
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    outliers = ((df[col] < q1 - 1.5 * iqr) | (df[col] > q3 + 1.5 * iqr)).sum()
    print(f'{col}: Median = {median:.2f}, IQR = {iqr:.2f}, Outliers = {outliers}')
```



Thống kê Median, IQR và số lượng Outliers:

fixed\_acidity: Median = 7.90, IQR = 2.10, Outliers = 41  
 volatile\_acidity: Median = 0.52, IQR = 0.25, Outliers = 19  
 citric\_acid: Median = 0.26, IQR = 0.34, Outliers = 1  
 residual\_sugar: Median = 2.20, IQR = 0.70, Outliers = 126  
 chlorides: Median = 0.08, IQR = 0.02, Outliers = 87  
 free\_sulfur\_dioxide: Median = 14.00, IQR = 14.00, Outliers = 26  
 total\_sulfur\_dioxide: Median = 38.00, IQR = 41.00, Outliers = 45  
 density: Median = 1.00, IQR = 0.00, Outliers = 35  
 pH: Median = 3.31, IQR = 0.19, Outliers = 28  
 sulphates: Median = 0.62, IQR = 0.18, Outliers = 55  
 alcohol: Median = 10.20, IQR = 1.60, Outliers = 12

### Chi tiết từng biến

- **fixed\_acidity:** Median = 7.9; IQR = 2.1; Outliers = 41 → phân bố tập trung, có một số giá trị acid cố định cao/thấp bất thường.
- **volatile\_acidity:** Median = 0.52; IQR = 0.25; Outliers = 19 → khá gọn, ít ngoại lai.
- **citric\_acid:** Median = 0.26; IQR = 0.34; Outliers = 1 → ổn định, gần như không có ngoại lai.
- **residual\_sugar:** Median = 2.2; IQR = 0.7; Outliers = 126 → đuôi dài, nhiều mẫu đường dư khác thường.
- **chlorides:** Median = 0.08; IQR = 0.02; Outliers = 87 → nhiều mẫu muối cao bất thường.
- **free\_sulfur\_dioxide:** Median = 14.0; IQR = 14.0; Outliers = 26 → phân tán mạnh, có điểm SO<sub>2</sub> tự do cao bất thường.
- **total\_sulfur\_dioxide:** Median = 38.0; IQR = 41.0; Outliers = 45 → SO<sub>2</sub> tổng phân tán mạnh.
- **density:** Median = 1.00; IQR ≈ 0.00; Outliers = 35 → mật độ hầu như sát 1, nhưng có vài điểm khác biệt.

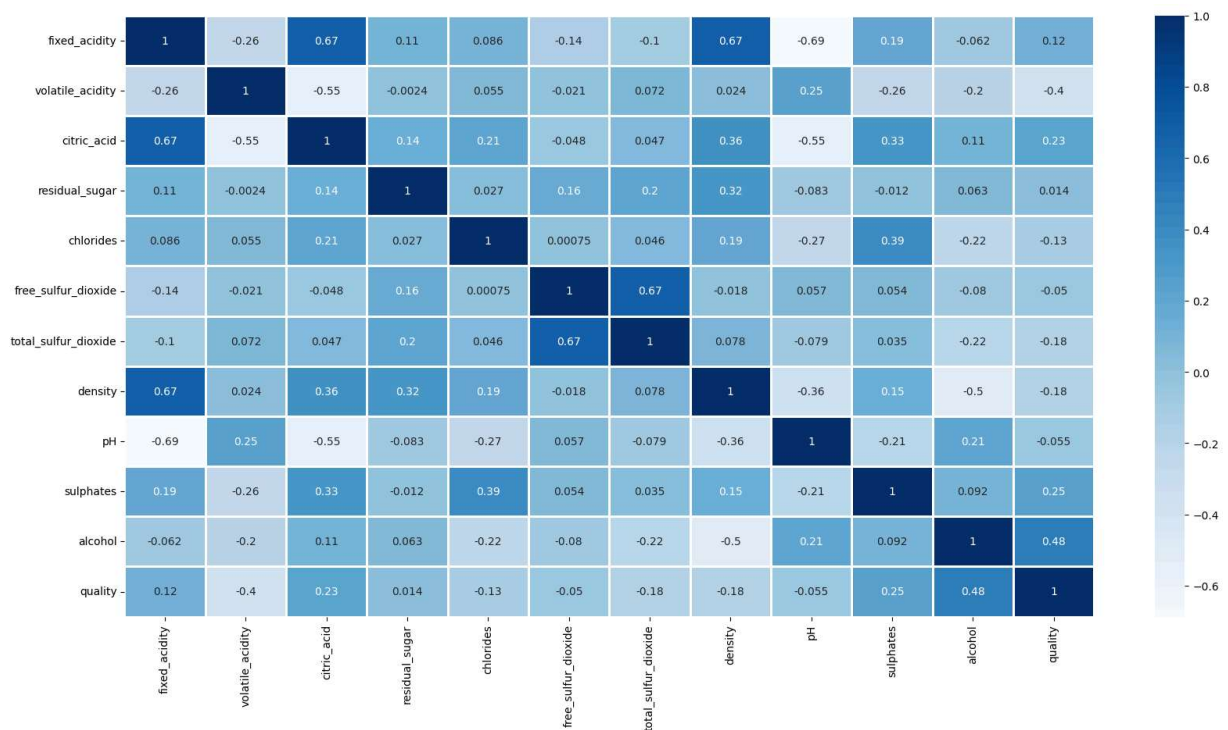
- **pH:** Median = 3.31; IQR = 0.19; Outliers = 28 → pH ổn định, ít ngoại lai.
- **sulphates:** Median = 0.62; IQR = 0.18; Outliers = 55 → vẫn có nhiều mẫu sulphates cao.
- **alcohol:** Median = 10.2; IQR = 1.6; Outliers = 12 → tập trung, ít ngoại lai.

**\*\*Kết luận \*\***

- Phần lớn biến hóa học của rượu vang **khá tập trung (IQR nhỏ)**.
- Một số biến có **nhiều ngoại lai** như `residual_sugar`, `chlorides`, `total_sulfur_dioxide`.
- Khi phân tích hoặc xây dựng mô hình nên cân nhắc **xử lý ngoại lai**

## 4.5 Correlation (Heatmap)

```
In [75]: plt.figure(figsize = [20, 10], facecolor = 'white')
sns.heatmap(df.corr(), annot = True, linewidths = 2, cmap = "Blues");
```



'Free sulfur dioxide' và 'Total sulfur dioxide' có tương quan dương khá cao ( $r \approx 0.67$ ).

'pH' và 'Fixed acidity' có tương quan âm khá cao ( $r \approx -0.68$ ).

Một số cặp biến khác có mức tương quan khoảng 0.5 (trung bình–khá).